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Explainable hybrid CNN-vision transformer with morphometric and cross-modal attention for simultaneous classification of rice seed ecology and variety

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ABSTRACT

Accurate classification of rice seed varieties and ecological classes is essential for seed quality assessment and precision agriculture. This paper proposes an explainable hybrid CNN–Vision Transformer (ViT) framework that combines data augmentation, handcrafted morphometric features, and cross-modal attention for the simultaneous classification of rice seed variety and rice ecology. A dataset comprising 16,000 high-resolution rice seed images collected from multiple agro-ecological zones and representing five rice varieties and three ecological classes, was used alongside biologically meaningful morphometric descriptors capturing seed size, shape, and structural characteristics. Experimental results demonstrate that data augmentation significantly enhances model generalization (2–4% accuracy improvement), while morphometric features provide consistent additional gains (1–3%). The inclusion of attention mechanisms further refines performance by focusing on discriminative seed regions. Among the evaluated models, the proposed smart attention-based hybrid CNN–ViT architecture, integrated with morphometric features and data augmentation, achieved superior performance for rice seed classification. The model obtained 96 ± 0.02% accuracy, 96% precision, 95% recall, and 95% F1-score for single-task variety classification, and 93 ± 0.02% accuracy with 93% precision, recall, and F1-score for single-task ecology classification. In the joint dual-task setting, the model achieved 90 ± 0.03% accuracy, with precision, recall, and F1-score all at 90%, indicating balanced performance across metrics. Furthermore, t-statistic and p-value significance tests were performed to confirm the statistical reliability of the performance improvements. To improve model transparency, Grad-CAM and attention heatmaps are employed to visualize the contribution of both visual and morphometric features, highlighting biologically relevant seed regions that influence predictions. The Explainability analysis confirms the interpretability and reliability of the proposed framework, supporting its applicability in real-world agricultural decision-making.

1. Introduction

Rice (Oryza sativa) is a staple food crop for more than half the world's population and is grown in an area of 150 million hectares. Rice is one of the most important cereal crops that was introduced to Ethiopia in the late 1970s and has since been growing in cultivation particularly in the areas with suitable climatic conditions. It is an emerging crop and economically important crop in Ethiopia [1].

The national rice research and development released 35 rice varieties for three ecosystems [1,2]. Ensuring the correct identification of rice seed varieties and their suitability for specific agro-ecological zones is crucial for improving crop productivity, strengthening food security, and enabling successful seed certification programs [2]. In many rice-producing countries, including those in Africa and Asia, farmers often face reduced yield due to the incorrect selection of seeds for the target environment [3]. Therefore, an intelligent and automated system capable of accurately differentiating rice ecologies such as upland, lowland, and irrigated growing systems alongside variety recognition, can support practical decision-making in precision agriculture.

Ecological adaptation of rice seeds is the result of long-term breeding

Abbreviations: AI, Artificial Intelligence; ANN, Artificial Neural Network; BPNN, Back Propagation Neural Network; CNN, Convolutional Neural Network; DIP, Digital Image Processing; GLCM, Gray Level Co-occurrence Matrix; Grad-CAM, Gradient-Weighted Class Activation Mapping; HOG, Histogram of Oriented Gradient; ILSVRC, ImageNet Large Scale Visual Recognition Challenge; KNN, K-Nearest Neighbors; PCA, Principal Component Analysis; ReLU, Rectified Linear Activation Function; RGB, Red, Green, Blue; ROI, Region of Interest; SURF, Speeded Up Robust Features; SVM, Support Vector Machine; ViT, Vision Transformer.

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practices designed to optimize plant performance under specific environmental conditions including water availability, soil structure, and climate characteristics [3]. These adaptations are expressed through measurable phenotypic variations in seeds such as length, width, shape index, hull texture, and coat pigmentation. As a result, agro-ecological categorization can be effectively inferred from seed morphology and appearance patterns, without requiring direct measurement of soil nutrients or temperature conditions [4]. This establishes a strong basis for visual and morphometric learning approaches to support ecology-related identification tasks [5,6].

Convolutional Neural Networks (CNNs) are commonly employed in image classification tasks because of their capacity to learn hierarchical spatial information from images. Recent improvements in deep learning have enabled CNN-based models to efficiently facilitate fast and automated seed trait analysis [7–10]. However, CNN approaches may struggle when seed varieties exhibit high visual similarity, which can lead to misclassification due to limited discriminative visual cues [11]. To address this limitation, morphological features such as area, perimeter, roundness, and aspect ratio can provide complementary agronomic information. These handcrafted descriptors enhance the feature representations learned by CNNs by incorporating structural and shape-based characteristics, thereby improving classification performance in visually similar seed categories [12–14].

Recently, Vision Transformers (ViTs) have emerged as a powerful alternative to CNN architectures due to their ability to capture long-range spatial dependencies through self-attention mechanisms. Since ecological and varietal features are frequently represented through modest global shape differences rather than just local textures, ViTs are especially useful for seed phenotyping. ViTs do not specifically optimize the interaction between handcrafted morphometric descriptors and deep visual representations when fused, despite incorporating internal attention for visual token learning. We introduce an extra post-fusion attention module that adaptively highlights the most informative modality and characteristics for each classification task in order to overcome this difficulty. By highlighting the relative significance of morphological and appearance signals in ecology and variety prediction, this cross-modal attention increases Explainability and discriminability between visually identical seeds.

Attention mechanisms have emerged as effective modules that improve feature learning by directing the model to concentrate on the most discriminative seed patches in order to overcome these constraints. Additionally, Explainability has grown in importance in agricultural AI, guaranteeing that agronomists and seed specialists can understand and rely on conclusions [15–21].

In this study, we proposed Hybrid CNN–Vision Transformer Framework that fuses deep visual representations with handcrafted morphometric descriptors for dual-task classification of rice seed ecology and variety. The model employs a feature-level fusion strategy and an attention module to improve class separability, while saliency-based visual explanations (Grad-CAM) provide transparency regarding which seed traits influence classification decisions [22]. Performance is comprehensively evaluated against baseline single-input and single-task architectures to demonstrate the advantages of feature integration and multi-task learning.

Unlike previous studies that either classify rice varieties using CNN-based visual features alone or rely solely on handcrafted morphometric descriptors, this work introduces a hybrid CNN–Vision Transformer framework enhanced with morphometric features and designed for dual-task learning. The key novelties of this study are:

1. **Integration of local and global visual features with morphometric descriptors:** CNNs capture fine-grained, local seed traits, while ViT encodes global contextual information, and these are fused with quantitative morphometric features within a unified architecture.
2. **Cross-modal attention fusion:** The model adaptively weighs complementary information from visual and morphometric modalities, improving class separability for both variety and ecological type.
3. **Simultaneous dual-task classification:** The framework jointly predicts rice variety and ecological adaptation in a single model, which has not been addressed in prior rice seed identification studies.
4. **Explainable attention visualization:** Grad-CAM and attention maps highlight biologically meaningful seed traits, linking model decisions to interpretable features.

Collectively, these innovations advance robust, explainable, and automated rice seed identification, supporting precision agriculture and intelligent seed selection.

2. Related works

Several studies were conducted focused on identifying rice leaf disease, type, and quality concerned with foreign rice using different techniques. The detail of each researcher contribution depicted in the following subsection part of the related work.

2.1. Rice seed identification methods

Early studies on rice seed classification mainly relied on traditional machine learning and handcrafted feature extraction techniques. Researchers used morphological, shape, and color features of rice seeds and trained models such as K-Nearest Neighbor (KNN), Decision Tree, Logistic Regression, Multilayer Perceptron (MLP), Random Forest (RF), and Support Vector Machine (SVM). Using 75,000 grains belonging to five rice varieties, the highest accuracy achieved was 89.91 % using MLP. However, these approaches depend on handcrafted features, which often fail to capture deep visual characteristics of seeds, leading to feature loss and reduced accuracy [23].

Other studies incorporated color, shape, texture, GIST, and SIFT features combined with ML classifiers including KNN, SVM, and RF, achieving a maximum accuracy of 90.54 % using RF. Despite improved performance, these methods still relied on manual feature extraction and did not consider the ecological traits of rice seeds, limiting their generalization capability [24].

Deep learning has been introduced for rice seed variety identification [25], showing improved performance over traditional ML. However, previous deep learning studies mostly focused on variety classification only, without addressing multi-trait (e.g., ecology) recognition.

2.2. Limitations of existing rice seed identification studies

- **Dependence on Handcrafted Features**

Traditional machine learning approaches rely heavily on manually engineered morphological, color, and texture descriptors, which often fail to capture subtle visual and structural variations in rice seeds. This results in loss of discriminative features and limits model robustness.

- **Absence of Multi-Trait Classification Frameworks**

Most prior works perform single-task identification, primarily focusing on variety classification. Ecological traits such as seed ecology, adaptability, and growing environment are not jointly modeled, leading to incomplete representation of seed characteristics.

- **Limited Dataset Diversity and Geographic Representation**

Existing studies are largely based on small, homogeneous datasets collected outside Ethiopia. The lack of diverse and region-specific samples restricts the generalizability and applicability of current

models to broader or localized agricultural settings.

- **Insufficient Use of Deep Learning for Multi-Modal or Multi-Task Learning**

Although deep learning has been introduced, prior work mainly employs single-task CNN architectures without integrating morphometric features or cross-modal information. This limits their capacity to learn comprehensive visual–structural relationships in rice seeds.

- **Lack of Explainability and Model Transparency**

Existing deep learning models often function as black boxes, offering no interpretable explanations for decision-making. This reduces trust and hinders adoption in agricultural research and seed certification workflows, where explainable reasoning is critical.

2.3. Rice leaf disease identification methods

Research efforts have also expanded into rice leaf disease diagnosis using ML, DL, and hybrid deep feature–based models. Traditional ML studies extracted handcrafted features and applied SVM for multi-class classification of three leaf diseases (Bacterial Leaf Blight, Brown Spot, and Leaf Smut), achieving 93.33 % training accuracy but only 73.33 % testing accuracy. The performance was limited due to insufficient noise removal, limited classifiers, and feature extraction inefficiency [26].

Another study used Random Forest, Decision Tree, Gradient Boosting, and Naïve Bayes on 600 images, with the best accuracy of 69.44 % using RF. The model performance was constrained by small dataset size and handcrafted feature dependence [27].

Deep learning methods improved disease identification accuracy. For instance, a CNN-based model trained on 1677 images classified 12 rice diseases with 96.5 % validation accuracy and 95.3 % test accuracy, but the dataset and class size were limited [7]. Transfer learning models such as AlexNet and VGG16 also achieved good performance (up to 93.11 %) on seven disease classes but suffered from insufficient dataset size and lack of image noise removal [8,9].

Hybrid approaches integrating deep features with ML have also been explored. Deep feature extraction from AlexNet, VGG16, VGG19, and ResNet50 combined with SVM classification yielded F1-scores up to 0.9838. Such models reduced training time but faced misclassification due to color similarity and data imbalance [28]. Similar works used CNN feature extraction with SVM for nine rice diseases, achieving 97.5 % accuracy, but dataset size and preprocessing limitations remained [29].

2.4. Limitations of existing studies

Despite notable progress in rice seed and leaf disease classification, existing methods still face several challenges. Most traditional models rely on handcrafted morphological, color, and texture features, which inadequately capture complex visual and structural variations. Deep learning approaches, though more effective, often focus solely on single-trait (variety) recognition and overlook ecological characteristics, limiting their practical utility. In addition, small and imbalanced datasets, insufficient noise handling, and lack of cross-modal feature integration constrain generalization. Furthermore, current deep models largely operate as black boxes without Explainability mechanisms, reducing interpretability and reliability in agricultural applications.

To address the above limitations, this study proposes an explainable hybrid transformer framework integrating morphometric feature fusion and cross-modal attention for dual-task classification of rice seed variety and ecology.

2.5. Explainable deep learning in agriculture

Recently, deep learning has been used alongside explainable artificial intelligence (XAI) to provide transparency in agricultural imaging tasks. XAI techniques such as Grad-CAM, LIME, SHAP, and attention visualization have been applied to plant disease detection to highlight the regions influencing model predictions. These methods increase farmer trust, support agronomists, and improve decision-making.

However, existing XAI applications have largely focused on plant disease recognition—not on rice seed identification, and none explore multi-trait rice seed ecology classification. Existing rice-related models classify either seed variety or leaf disease but do not provide feature-level Explainability to interpret ecological and phenotypic traits of rice seeds.

2.6. Summary of gaps and research need

Existing studies on rice seed classification and rice plant health analysis have demonstrated the effectiveness of both handcrafted and deep learning–based approaches. However, several critical gaps remain unaddressed. Current methods predominantly focus on single-trait recognition, mainly variety classification, without jointly considering ecological attributes that are essential for breeding and regional adaptability. Most models rely exclusively on visual features and do not incorporate complementary morphometric descriptors, resulting in limited discriminative capability. Additionally, challenges such as small and imbalanced datasets, noise sensitivity, and high intra-class similarity continue to restrict generalization performance.

Furthermore, the lack of Explainability in deep models limits their acceptance in agricultural decision-making contexts where model transparency is crucial. These gaps highlight the need for a comprehensive and interpretable framework capable of integrating multi-modal features and performing dual-task classification to enhance both performance and practical applicability.

In Table 1 the summary of research gaps and research needs are illustrated.

3. Materials and methods

The proposed model introduces an Explainable Attention-Based Vision Transformer (ViT) framework for simultaneous classification of rice seed ecology and variety. The model aims to address the limitations of traditional handcrafted feature extraction methods by leveraging deep feature representations and attention mechanisms that can learn intrinsic seed ecology and variety. The proposed model architecture illustrated in Fig. 1.

- **Input Image Module** – acquisition and preprocessing of rice seed images.

Table 1
The summary of research gaps and research.

Identified Gaps in Existing Studies	Research Needs / Opportunities
Focus mainly on single-trait classification, especially variety identification	Develop models that jointly classify multiple traits (e.g., variety and ecology) in a unified framework
Reliance solely on handcrafted or visual features	Integrate visual and morphometric features for richer and more discriminative representation
Limited dataset diversity, class imbalance, and noise sensitivity	Enhance robustness through improved feature learning, augmentation, and preprocessing strategies
Lack of cross-modal feature interaction in current deep models	Introduce cross-modal attention mechanisms for better feature fusion and contextual learning
Deep learning models used as black boxes without interpretability	Incorporate Explainability tools (e.g., attention visualization) to improve transparency and trust
Separate models needed for different classification tasks	Employ multi-task learning to reduce computational cost and improve prediction consistency

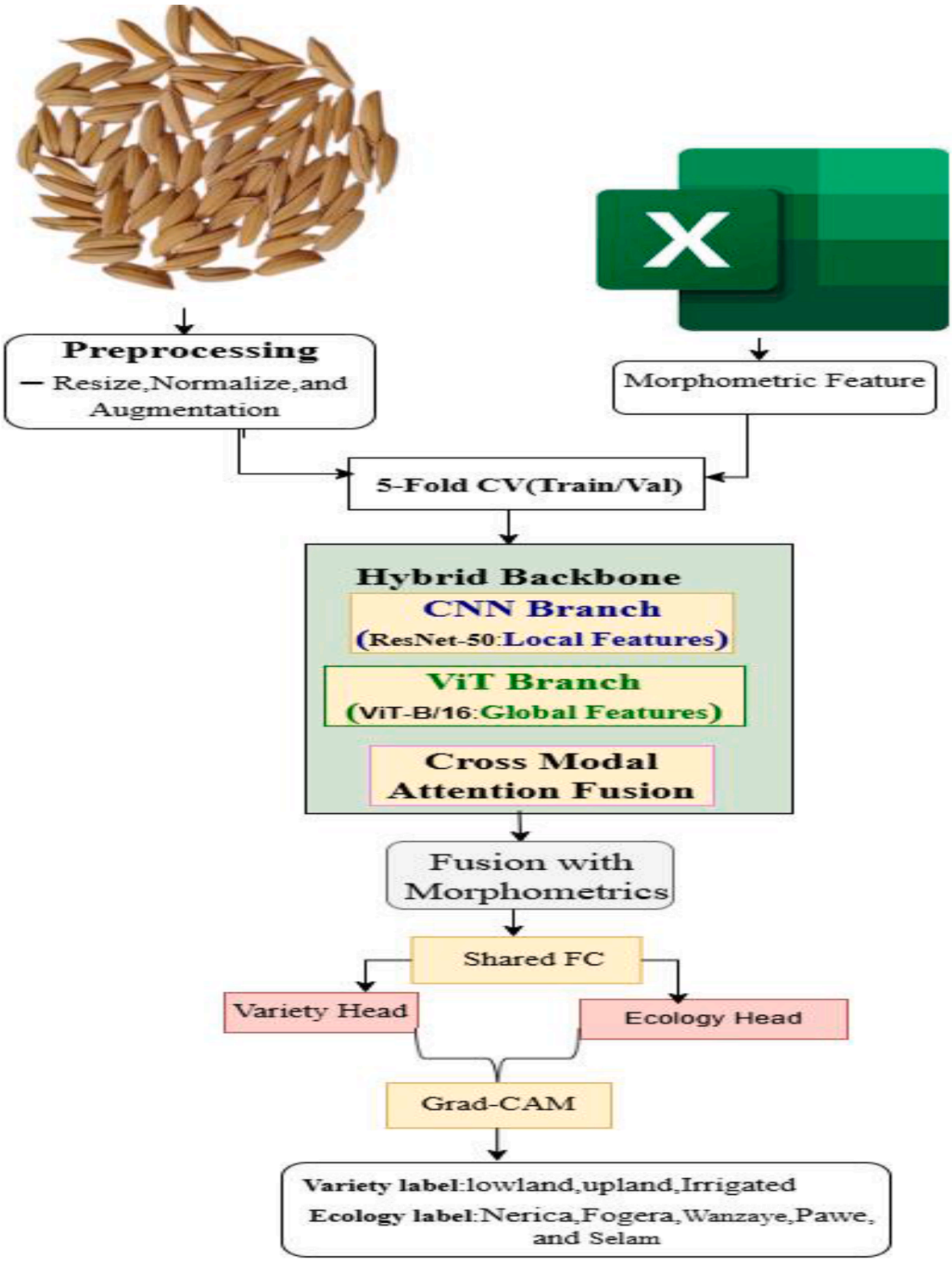


Fig. 1. The proposed architecture.

- **Morphometric Feature Extraction:** Morphological descriptors including: length, width, area, perimeter, aspect ratio, roundness, eccentricity, and solidity.
- **CNN:** extract local features of the images.
- **Vision Transformer:** accept input images and extract global visual feature vectors.
- **Feature Fusion with Attention**: Visual and morphometric features are concatenated.
  - The attention layer learns the relative importance of each feature.
- **Classification Head –** Ecological and Variety Classification.

3.1. Dataset description and preparation

The dataset used in this study comprises high-resolution rice seed images collected from multiple agro-ecological zones, along with corresponding morphometric features. The morphometric descriptors include length, width, area, perimeter, aspect ratio, roundness, eccentricity, and solidity, capturing biologically relevant traits that complement visual information. A total of 16,000 rice seed images were included, categorized into three ecological classes (upland, lowland, and irrigated) and five rice seed varieties. All images were acquired from the

Pawe Agricultural Research Institute, ensuring consistent imaging conditions for accurate phenotypic analysis. Fig. 2 illustrates representative samples of seeds from the three ecological categories, highlighting the subtle morphological differences that inform both ecology and variety classification.

The rice seeds in the dataset also belong to five distinct varieties: Nerica, Fogera, Pawe, Wanzaye, and Selam. Representative images of each variety are shown in Fig. 3, illustrating the subtle visual and morphological differences used for variety classification.

To enhance the quality and representation of the dataset, a series of preprocessing and augmentation techniques were applied. Preprocessing involved image resizing to a uniform resolution suitable for Vision Transformers, normalization to ensure consistency across the dataset [30–43].

Data augmentation techniques such as random rotation, flipping, zooming, and contrast adjustments were employed to increase dataset diversity and prevent overfitting [44–53].

The dataset was divided into 5 cross fold validation subsets using a stratified split to maintain class distribution balance.

A controlled data augmentation method was employed to improve model generalization and reduce overfitting. A balanced dataset of 3200 photos was produced by artificially using the Keras ImageDataGenerator function. To maintain seed boundary features, fill_mode='nearest' was used to apply the chosen augmentations, which comprised rotation (±30°), horizontal and vertical flipping, width and height shifting (±10 %), and zooming (±10 %).

Because they replicate realistic changes in seed orientation, imaging viewpoint and camera distance without producing geometric deformation or changing fundamental morphometric features like length, width, and aspect ratio, these augmentation techniques were specifically selected. This maintained the biological validity of both shape-based measures and visual deep learning features. Furthermore, in order to preserve the integrity of trait-dependent seed categorization, distortion-inducing transformations like shear and elastic deformations were eliminated. Overall, the used augmentation increased robustness and guaranteed that the suggested model could correctly identify rice seeds in a variety of real-world imaging scenarios. The sample augmented images illustrated in Fig. 4.

3.2. Cross-Validation

To ensure robust and unbiased evaluation, a 5-fold stratified cross-validation strategy was employed. The dataset was divided into five equal subsets, maintaining the original class distribution of both rice seed varieties and ecological types. The model was iteratively trained on four folds and tested on the remaining fold, with performance metrics averaged across all folds to assess generalization and stability. The primary dataset consists of rice seed images from five Ethiopian varieties under different ecological conditions. While most varieties are primarily associated with a single ecology, a subset of images includes varieties under multiple ecological conditions to enable dual-task learning. An independent dataset collected from the Fogera Research Center was used to validate generalization, where the same varieties appear under multiple ecological conditions.

Table 2 summarizes the dataset composition before and after the application of data augmentation for both training and testing folds.

Each rice variety appears across multiple ecological conditions. Therefore, ecology cannot be inferred directly from the variety label. This design ensures that the dual-task learning framework learns independent visual cues for both variety identification and ecology classification, preventing trivial prediction.

3.3. Morphometric feature extraction

To capture biologically relevant characteristics of rice seeds, handcrafted morphometric descriptors were extracted from each seed image using standard image processing techniques. These features quantify the size, shape, and surface properties of seeds and provide complementary information to visual deep learning features.

Shape Features included:

- **Length** and **width**: measured along the major and minor axes of the seed.
- **Area** and **perimeter**: capturing the overall size and boundary length.
- **Aspect ratio**: the ratio of length to width, indicating elongation.
- **Roundness** and **eccentricity**: describing the seed's geometric shape.
- **Solidity**: the proportion of the seed area relative to its convex hull, reflecting compactness.

All extracted features were normalized using Min-Max scaling to

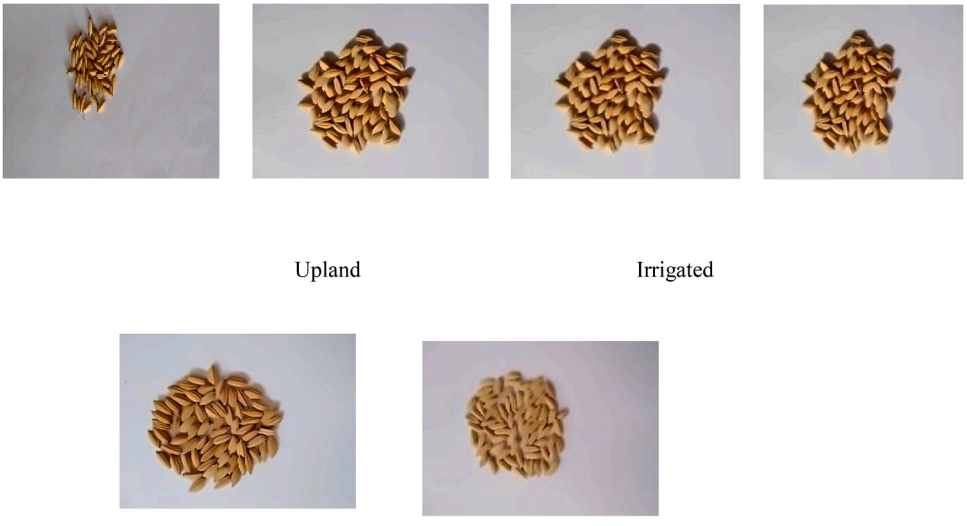


Fig. 2. The sample seed images for ecological types.

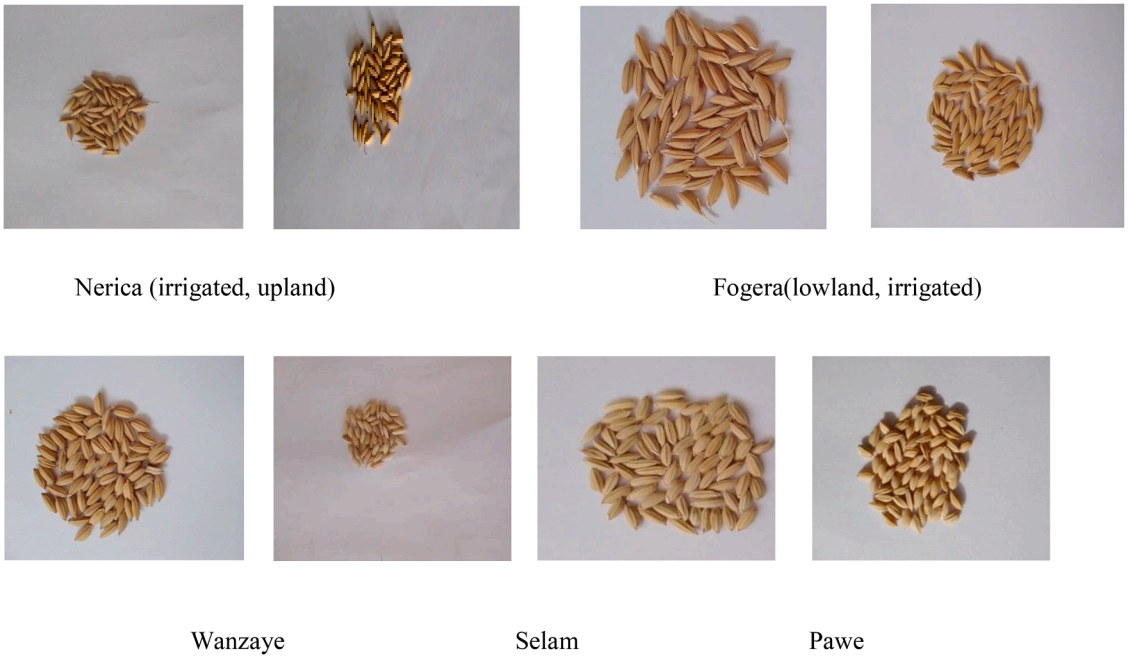


Fig. 3. The sample rice seed varieties.

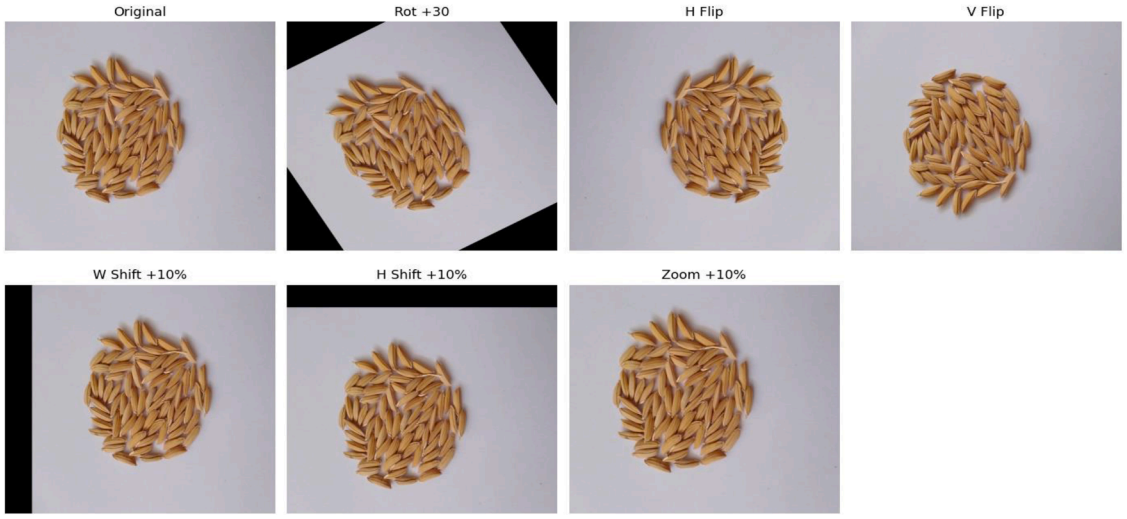


Fig. 4. Sample output augmented images (Rotation, Flip, Width shift, Height shift, Zoom, respectively).

Table 2
The overall dataset description.

Variety	Ecology	Number of images	Test per fold (20 %)	Train Per fold (80 %)	After-Aug	Test per fold (20 %)	Train Per fold (80 %)
Nerica	Irrigated	2000	400	1600	2320	464	1856
Nerica	Upland	2000	400	1600	2320	464	1856
Selam	Upland	2000	400	1600	2640	528	2112
Fogera	Lowland	2000	400	1600	2320	464	1856
Fogera	Irrigated	2000	400	1600	2320	464	1856
Wanzaye	Lowland	2000	400	1600	2640	528	2112
Pawe	Upland	2000	400	1600	2320	464	1856
Pawe	Lowland	2000	400	1600	2320	464	1856
Total		16,000	3200	12,800	19,200	3840	15,360

ensure numerical stability and compatibility with the deep learning model. This preprocessing step allows the morphometric descriptors to be effectively combined with the high-dimensional features extracted from the Vision Transformer, facilitating hybrid feature fusion for dual-task classification.

3.4. Convolutional neural network (CNN) branch

The CNN branch utilizes a ResNet-50 backbone, pretrained on ImageNet, to extract hierarchical local visual features from rice seed images [11,15–21,54–60]. The network consists of a convolutional stem, four residual blocks, and a global average pooling layer. The output feature maps are projected into a feature vector that is subsequently fused with ViT features via a cross-modal attention module. This branch captures fine-grained textures, seed tips, and shape patterns that complement the global context captured by the ViT branch.

3.5. Vision transformer (ViT) branch

In order to extract global visual characteristics from rice seed images, long-range spatial correlations that are frequently overlooked by convolutional neural networks of the Vision Transformer (ViT) branch were utilized. ViT considers a picture as a series of non-overlapping patches that are linearly embedded and concatenated with positional encodings to maintain spatial information, in contrast to traditional CNNs.

Fixed-size patches were initially created from each seed image, flattened, and then projected into a high-dimensional feature space. The transformer encoder's multi-head self-attention layers processed this patch embedding, enabling the model to identify correlations between far-off areas of the seed image, including surface patterns, texture, and shape. The ViT branch's output is a dense feature vector that highlights each seed's key visual traits.

In order to take advantage of transfer learning, improve feature representation, and shorten training times, the ViT branch was started with pretrained weights from ImageNet. To increase model stability and avoid overfitting, dropout and layer normalization were used. The cross-modal attention module then combines these visual characteristics with morphometric descriptors to allow dual-task classification of rice seed variety and ecosystem.

3.6. Morphometric feature branch

Each rice seed's hand-crafted numerical descriptors, such as length, breadth, area, perimeter, aspect ratio, roundness, eccentricity, and solidity, are processed by the morphometric feature branch. In addition to the visual representations derived from the CNN and ViT branch, these features offer biologically interpretable information.

To guarantee numerical stability and compatibility with the deep learning system, the morphometric feature vector was first normalized using Min-Max scaling. Following normalization, the features underwent batch normalization and a sequence of fully linked (dense) layers using ReLU activation functions. This creates a high-level feature vector that can be fused with the visual features from the low-dimensional numerical data.

Dropout layers were added after each thick layer to avoid overfitting because of the tiny sample size. Essential shape and size information is captured in the morphometric branch's output in a format that can be successfully combined with the visual features obtained by CNN and ViT in the feature fusion and cross-modal attention module that follows.

The model can accurately classify the ecology and variety of rice seeds by using both precise morphometric measures and global visual patterns thanks to its hybrid representation.

3.7. Feature fusion layer and cross-modal attention

A feature fusion approach was used to efficiently integrate the complimentary data from the morphometric descriptors and the CNN, and Vision Transformer (ViT) visual features. The high-dimensional visual feature vector F_{vision} from the CNN, and ViT branch and the dense morphometric feature vector F_{morph} were concatenated to form a unified representation F_{fusion}. This hybrid feature vector provides a richer representation for dual-task classification by combining biologically interpretable seed properties with global picture attributes. To enhance the model's ability to selectively focus on the most relevant features from each modality, a cross-modal attention mechanism was applied to F_{fusion}.

$$F_morph = W_m * F_morph + b_m \; Feature \; dimension \tag{1}$$

$$Q = W_Q F_{fusion}, \; K = W_K F_{fusion}, \; V = W_V F_{fusion}, \; Attention(Q, K, V) = Softmax\left(\backslash frac\{Q K^T\}\left\{\sqrt{\{d_k\}}\right\}\right) V \; Cross - modal \; attention \; calculation \; for \; fused \; features \tag{2}$$

Where:

- F_{fusion} = concatenated feature vector from CNN/ViT visual embedding and projected morphometric features
- W_Q, W_K, = learnable projection matrices
- d_k = dimension of the key vectors
- Softmax (·) = normalizes attention scores so that weights sum to 1

This module learns task-specific attention weights, allowing the model to adaptively emphasize important visual regions and morphometric features for each classification task (ecology and variety). The attention scores are computed using scaled dot-product attention, similar to standard transformer self-attention, but applied across the concatenated modalities.

This approach enables the model to:

1. Dynamically weigh the contribution of each feature type (visual vs. morphometric)
2. Capture subtle interactions between seed morphology and visual patterns that are predictive of ecology or variety
3. Improve interpretability, as the attention weights can be visualized to identify which features influence each task

The output of the cross-modal attention module, a refined fused feature vector, is then fed into two task-specific heads for simultaneous prediction of rice seed ecology and variety. This design ensures that both tasks benefit from shared feature representations while allowing the model to focus on modality-specific cues critical for accurate dual-task classification.

The overall module structure and information flow for the hybrid CNN–ViT + morphometric + attention framework is illustrated in Fig. 5. The schematic highlights the CNN/ViT visual feature branch, morphometric feature projection, feature fusion, attention weighting, and final classification for rice seed variety and ecological type.

3.8. Dual-Task output heads

To enable simultaneous dual-task categorization, the cross-modal attention module's fused feature vector is routed into two distinct task-specific output heads:

Ecology Head:

1) Predicts the ecological category of each rice seed (upland, lowland, irrigated)
2) Composed of fully connected layers followed by a softmax activation, generating class probabilities.

Variety Head:

1) Predicts the rice seed variety (Nerica, Fogera, Pawe, Selam, and Wanzaye)
2) Similarly structured with dense layers and a Softmax output layer.

Each head is trained independently but shares the common fused

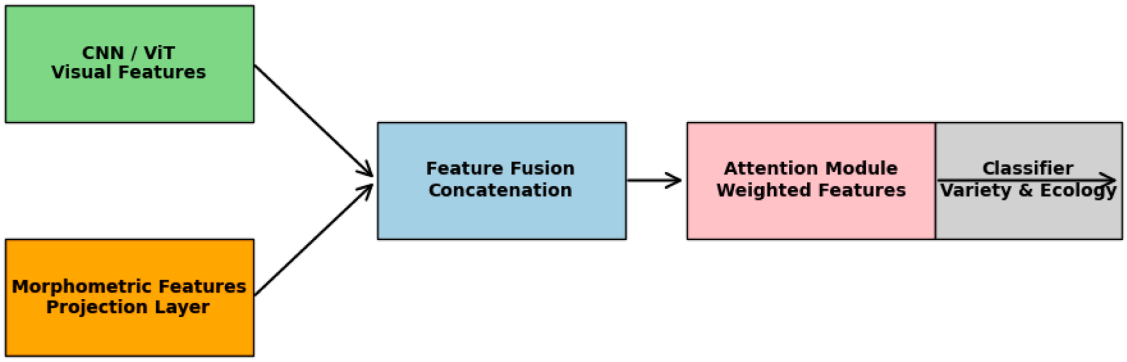


Fig. 5. Schematic of the hybrid CNN–ViT + Morphometric + Attention module.

features, enabling multi-task learning that leverages correlations between the two tasks.

Loss Function:

The total training loss is defined as a weighted sum of the individual task losses:

$$Ltotal = \lambda 1 Lecology + \lambda 2 Lvariety \text{ The training loss} \quad (3)$$

Where, Lecology and Lvariety are categorical cross-entropy losses computed for each head and λ1 and λ2 weighting factors used to balance the influence of each task (typically λ1=λ2=0.5for equal contribution)

This dual-head architecture allows the model to jointly optimize for both ecological type and variety predictions, while the cross-modal attention module ensures task-specific feature emphasis, improving accuracy and interpretability for both tasks.

3.9. Evaluation of the model

Standard measures such as accuracy, precision, recall, F1-score, and confusion matrices were used to assess the dual-task model independently for ecology and variety categorization. Five-fold stratified cross-validation was used for evaluation, and metrics were averaged over all folds to produce a reliable performance estimate. Standard measures such as accuracy, precision, recall, F1-score, and confusion matrices were used to assess the dual-task model independently for ecology and variety categorization. We also used macro-averaged metrics across both tasks. Five-fold stratified cross-validation was used for evaluation, and metrics were averaged over all folds to produce a reliable performance estimate.

3.10. Explainability module

To ensure model transparency and interpretability—critical for agricultural researchers, agronomists, and plant breeders the proposed architecture integrates Explainable AI (XAI) mechanisms.

Two major Explainability tools are employed:

- **Attention Maps** generated from the attention layers to visually highlight seed regions that contributed to the model prediction.
- **Gradient-weighted Class Activation Mapping (Grad-CAM)** applied on hybrid ViT backbones to interpret localized feature importance at earlier feature extraction levels.

Explainability assists domain experts in validating whether the model attends to biologically meaningful seed parts. This builds trust in AI-driven decision-making, supports scientific understanding, and encourages adoption of automated ecological and variety-based seed identification systems.

3.11. Experimental setup

The experimental evaluation was conducted using a high-performance Google Colab computing environment with an NVIDIA T4 GPU (16 GB). The proposed model was implemented using the TensorFlow deep learning framework, along with supporting libraries for data preprocessing, training, and visualization by using phyton version 3.9.

All hyperparameters used during training are summarized in Table 3. The CNN backbone uses a ResNet-50 architecture to extract hierarchical local feature representations, whilst the Vision Transformer (ViT-B/16) branch models global contextual dependencies via multi-head self-attention methods. The selected hyperparameters, including learning rate, batch size, and optimizer, were determined based on preliminary experiments to ensure stable convergence and optimal performance.

Input images were resized to 224 × 224 and normalized prior to training. Data augmentation techniques were applied only to the training set to improve model generalization.

Training was performed for 50 epochs using the Adam optimizer and categorical cross-entropy loss function. Early stopping and model check pointing were employed to prevent overfitting and retain the best-performing model.

Training was executed on a Google colab, and early stopping with model check pointing was applied to avoid overfitting. Performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix were computed for model evaluation.

Table 3
List of hyperparameters with their value.

Hyperparameters	Value
Input image size	224 × 224 × 3
Batch size	32
Learning rate	0.0001
Optimizer	Adam (β1 = 0.9, β2 = 0.999)
Epochs	50
Loss Function	Categorical Cross-Entropy
Validation strategy	5-fold cross-validation
Early stopping	Patience = 10 (monitor: validation loss)
CNN backbone	ResNet-50 (pretrained on ImageNet, ReLU activations)
Transformer branch	ViT-B/16: 12 encoder layers, patch size 16 × 16, hidden dim 768, 12 attention heads, MLP dim 3072, dropout =0.1
Hybrid Fusion	Multi-head attention fusion between ResNet-50 and ViT outputs; morphometric features concatenated
Preprocessing	Resize to 224 × 224, normalize; augmentation applied only during training
Data augmentation	Applied to training set only

4. Experimental results and discussion

4.1. Model performance overview

The proposed Explainable Hybrid Vision Transformer with Morphometric and Cross-Modal Attention was evaluated on the rice seed dataset to assess its ability to simultaneously classify seed variety and ecological zone. To ensure robustness and avoid bias, we employed 5-fold cross-validation. The dataset was partitioned into five folds, where in each iteration 80 % of the data were used for training and 20 % for testing. Performance metrics including precision, recall, F1-score, and accuracy are reported as mean ± standard deviation across the five folds. The model's performance was analyzed using standard classification metrics including accuracy, precision, recall, and F1-score.

- **Accuracy** — proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \; Accuracy\ metrics \tag{4}$$

- **Precision (per class)** — proportion of predicted positives that are true positives:

$$Precision = \frac{TP}{TP + FP} \; precision\ metrics \tag{5}$$

- **Recall (Sensitivity, per class)** — proportion of actual positives correctly identified:

$$Recall = \frac{TP}{TP + FN} \; Recall\ metrics \tag{6}$$

- **F1-score (per class)** — harmonic mean of precision and recall:

$$F1 - score = \frac{2.Precision.recall}{Precision + Recall} \; F1 - score\ metrics \tag{7}$$

- **Confusion matrix** — to show per-class classification counts and misclassification patterns.

TP: True Positive, TN: True Negative, FP: False Positive, FN: False Negative.

Precision measures the correctness of positive predictions, Recall measures coverage of actual positives, F1-score balances Precision and Recall.

All reported metrics are computed on the held-out test set. Macro-averaged metrics are reported when classes are balanced; otherwise, both macro and weighted averages are shown.

4.2. Quantitative results

This section presents the results of rice seed and rice ecology variety identification using CNN and Vision Transformer (ViT), both with and without data augmentation, combined with morphometric features, and their quantitative analyses. Comparisons are made to evaluate the effectiveness of each approach.

4.2.1. Rice seed variety identification

In the first experiment, the rice seed variety identification model was trained to classify five varieties, and the detailed dataset distribution is presented in Table 4.

The split is repeated across the five cross-validation folds.

Table 4
The data distribution for rice seed variety.

Variety	Total images	Test per fold (20 %)	Train Per fold (80 %)
Nerica	4000	800	3200
Selam	2000	400	1600
Fogera	4000	800	3200
Wanzaye	2000	400	1600
Pawe	4000	800	3200
Total	16,000	3200	12,800

4.3. CNN-based models

We first evaluated ResNet-50 CNN models under four configurations:

1. CNN on original images
2. CNN with data augmentation
3. CNN with morphometric feature fusion
4. Attention-based CNN with morphometric feature fusion

Performance improved consistently across these stages, demonstrating the benefit of augmentation and the complementary role of morphometric descriptors. Table 5 illustrates the class-wise performance of the ResNet-50 backbone CNN model, while Fig. 6 presents a joint visualization of prediction outcomes and class-wise evaluation metrics, including the confusion matrix, precision, recall, F1-score, and support, based on ordinal images.

The ResNet-50 backbone CNN model demonstrates strong classification performance with an overall accuracy of 87 %. As shown in Table 5, the Fogera class achieves the highest performance, with a precision, recall, and an F1-score of 89 %, indicating excellent class separability and minimal misclassification. Pawe also shows stable performance with balanced precision and recall values of 88 %, while Wanzaye and Selam records slightly lower recall (84 %), suggesting that although most Nerica, Fogera, and Pawe samples are correctly identified, some false positives remain. The confusion matrix further reveals that the primary source of misclassification occurs between the Selam and Wanzaye classes, where 64 Selam samples are incorrectly classified and 52 Wanzaye samples are misclassified, indicating visual similarity between these to the rest varieties. In contrast, Nerica, Fogera, and Pawe exhibit strong robustness, with only less samples misclassified as compared to the rest class, confirming its distinct morphological characteristics. Overall, these results highlight the effectiveness of the proposed CNN framework while identifying class overlap as the main challenge for further improvement.

4.4. CNN with data augmentation

This section presents the experimental results of the CNN model trained using data augmentation techniques to enhance model generalization and reduce overfitting. The performance of the proposed approach is evaluated using class-wise metrics, including precision, recall, F1-score, and support, along with the confusion matrix to provide a detailed assessment of classification behavior across the Selam, Wanzaye, Nerica, Fogera, and Pawe classes. Table 6 illustrates the class-wise

Table 5
The performance of the CNN for rice seed variety classification on original images.

Metric	Class					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	84	82	88	89	88	86±0.03
Recall (%)	84	87	86	89	88	87±0.02
F1-score (%)	84	84	87	89	88	86±0.02
Support (n)	400	400	800	800	800	3200
Accuracy (%)						87±0.02

	precision	recall	f1-score	support
Selam	0.84	0.84	0.84	400
Wanzaye	0.82	0.87	0.84	400
Nerica	0.88	0.86	0.87	800
Fogera	0.89	0.89	0.89	800
Pawe	0.88	0.88	0.88	800
accuracy			0.87	3200
macro avg	0.86	0.87	0.86	3200
weighted avg	0.87	0.87	0.87	3200

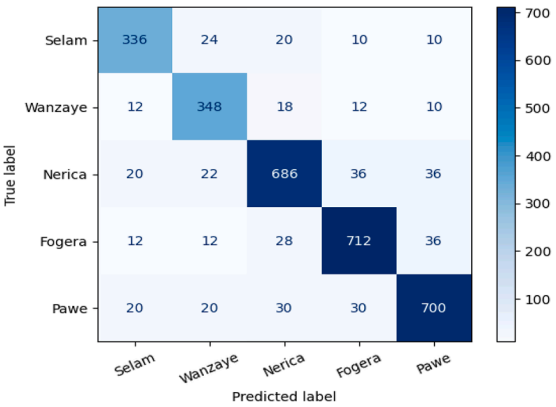


Fig. 6. Comprehensive evaluation of the proposed model showing the confusion matrix and class-wise performance metrics on the test dataset.

Table 6
The performance of the CNN for rice seed variety classification on augmented images.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	86	87	89	91	89	86±0.01
Recall (%)	87	86	91	87	91	87±0.03
F1-score (%)	86	87	90	89	90	86±0.03
Support (n)	528	528	928	928	928	3840
Accuracy (%)						89±0.03

performance of the CNN model, while Fig. 7 presents a joint visualization of prediction outcomes and class-wise evaluation metrics, including the confusion matrix.

The CNN model trained with data augmentation achieved an overall accuracy of 89 % (Table 6), showing improved performance. Among the five classes, Nerica and Pawe achieved the best results, with 89 % precision, 91 % recall, and 91 % F1-score, indicating strong class separability and effective feature learning. Fogera also performed well with 91 % precision and 87 % recall, while Wanzaye showed slightly lower precision (87 %) and recall (86 %), suggesting some misclassification.

The confusion matrix supports these results. Selam had 460 correctly classified samples, with 68 errors, mainly due to visual similarity with other varieties. Wanzaye recorded 453 correct predictions and 75 misclassifications, indicating class overlap. In contrast, Nerica, Fogera, and Pawe showed more robust classification, reflecting clearer morphological differences. Overall, the results demonstrate that data augmentation improves CNN's generalization and classification performance.

4.5. CNN with morphometric feature fusion

In this experiment, morphometric features were fused with deep CNN representations to capture complementary structural information and strengthen class discrimination. Table 7 summarizes the classification performance of the CNN model on augmented rice seed images incorporating morphometric feature fusion, whereas Fig. 8 depicts a comprehensive evaluation through the confusion matrix and class-wise performance metrics on the test set.

The CNN model with morphometric feature fusion achieved strong performance, reaching 90 % overall accuracy (Table 7). Pawe showed the best results with 90 % precision, 93 % recall, and 91 % F1-score, indicating clear class separability. Nerica also performed well with 91 % precision, 90 % recall, and 90 % F1-score. Selam achieved 86 % precision and 90 % recall, while Wanzaye recorded 88 % precision and 87 % recall, showing reliable detection with minor misclassification. The confusion matrix indicates 474 correctly classified Selam samples and 459 for Wanzaye, with some overlap between classes. Pawe remained highly distinct with 863 correct predictions. Overall, macro

Table 7
The performance of the CNN for rice seed variety classification on augmented images with morphometric.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	86	88	90	92	90	86±0.01
Recall (%)	90	87	91	87	93	87±0.02
F1-score (%)	88	88	90	89	91	86±0.01
Support (n)	528	528	928	928	928	3840
Accuracy (%)						90±0.02

	precision	recall	f1-score	support
Selam	0.86	0.87	0.86	528
Wanzaye	0.87	0.86	0.87	528
Nerica	0.89	0.91	0.90	928
Fogera	0.91	0.87	0.89	928
Pawe	0.89	0.91	0.90	928
accuracy			0.89	3840
macro avg	0.88	0.88	0.88	3840
weighted avg	0.89	0.89	0.89	3840

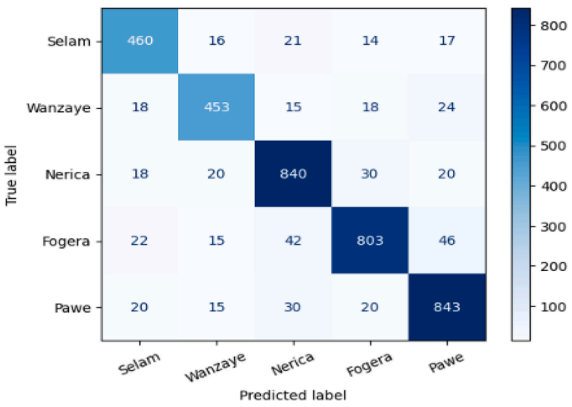


Fig. 7. Comprehensive evaluation of the proposed model showing the confusion matrix and class-wise performance metrics on the test dataset.

	precision	recall	f1-score	support
Selam	0.86	0.90	0.88	528
Wanzaye	0.88	0.87	0.88	528
Nerica	0.90	0.91	0.90	928
Fogera	0.92	0.87	0.89	928
Pawe	0.90	0.93	0.91	928
accuracy			0.90	3840
macro avg	0.89	0.89	0.89	3840
weighted avg	0.90	0.90	0.90	3840

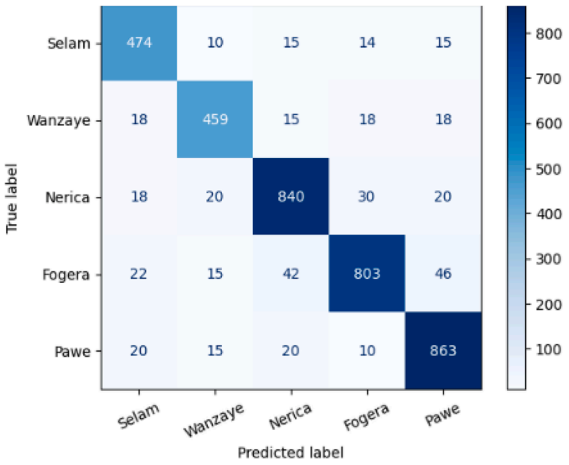


Fig. 8. Comprehensive evaluation of the proposed model showing the confusion matrix and class-wise performance metrics on the test dataset.

and weighted averages of 93 % demonstrate balanced performance, confirming that morphometric feature fusion significantly improves CNN classification reliability.

4.6. Attention-based CNN with morphometric feature fusion

In this study, we propose an attention-based CNN integrated with morphometric feature fusion to enhance feature representation and improve classification performance of rice seed varieties. Table 8 presents the class-wise precision, recall, F1-score, and overall accuracy of the model, while Fig. 9 illustrates the corresponding confusion matrix and detailed classification results.

The attention-based CNN with morphometric feature fusion achieved 92 % overall accuracy on the rice seed dataset (Table 8). Pawe showed the best performance with 92 % precision, 95 % recall, and 94 % F1-score, highlighting clear separability. Nerica also reached 92 % across all metrics, while Selam and Wanzaye recorded 89–91 % precision/recall, indicating reliable identification despite some overlap. The confusion matrix shows minor misclassifications—Selam had 481 correct predictions, Wanzaye 460, and Pawe remained highly distinct—confirming class-specific separability. Macro and weighted averages of 91 % demonstrate balanced performance. Overall, integrating attention mechanisms with morphometric features enhances CNN discriminative power, reduces errors, and improves rice seed classification reliability.

Table 9 summarizes the overall rice seed variety classification performance, including results from statistical tests

Table 9 shows the incremental improvements in classification accuracy as each module is added to the baseline CNN. Each addition — data augmentation, morphometric features, and the attention module contributes positively to model performance. All improvements are statistically significant ($p < 0.001$), confirming that each component meaningfully enhances the model's predictive capability. Notably, the attention module provides the largest performance boost, highlighting its critical role in capturing important spatial and feature-level information. Overall, this study demonstrates that the combined use of augmentation, morphometric features, and attention mechanisms produces a robust and highly accurate model for rice seed variety classification.

Table 8
The performance of the CNN for rice seed variety classification on augmented images with attention based morphometric.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	89	89	92	93	92	86±0.02
Recall (%)	91	87	92	90	95	87±0.02
F1-score (%)	90	88	92	91	94	86±0.02
Support (n)	528	528	928	928	928	3840
Accuracy (%)						92±0.03

4.7. ViT-based models

Vision Transformer models were evaluated using the same four configurations:

1. ViT on original images
2. ViT with data augmentation
3. ViT with morphometric feature fusion
4. Attention-based ViT with morphometric feature fusion

Results indicate that ViT benefits significantly from morphometric descriptors due to strong complementarity between global attention and explicit shape information.

5. ViT on original images

In this experiment, the Vision Transformer model was trained and evaluated on the raw/original rice seed images without applying any augmentation techniques. The purpose of this experiment was to investigate how effectively ViT can extract discriminative visual features for variety recognition using only the original image distribution. The precision, recall, and f1-score result of this experiment presented in Table 10. The visual representation of the confusion matrix output result illustrated in Fig. 10.

The confusion matrix further reveals important class-wise insights regarding the discriminative power of the Vision Transformer. The Pawe and Fogera variety exhibits the strongest class separability, with 744 and 716 out of 800 for each samples correctly classified, indicating that this class possesses more distinct morphological attributes—such as noticeable differences in shape, texture, or color features—making it easier for the model to recognize. In contrast, the misclassification occurs between Selam and Wanzaye, which led to less accuracy. This strong bidirectional confusion suggests that the two varieties share highly similar visual phenotypes, likely due to overlapping size, color, and pattern characteristics. There is also a degree of misclassification involving Nerica compared to the other two varieties strengthens the conclusion that Fogera and Pawe is visually more distinguishable from Selam, Nerica, and Wanzaye, which may require further augmentation or feature-enhancement strategies to improve class discrimination.

	precision	recall	f1-score	support
Selam	0.89	0.91	0.90	528
Wanzaye	0.89	0.87	0.88	528
Nerica	0.92	0.92	0.92	928
Fogera	0.93	0.90	0.91	928
Pawe	0.92	0.95	0.94	928
accuracy			0.92	3840
macro avg	0.91	0.91	0.91	3840
weighted avg	0.92	0.92	0.92	3840

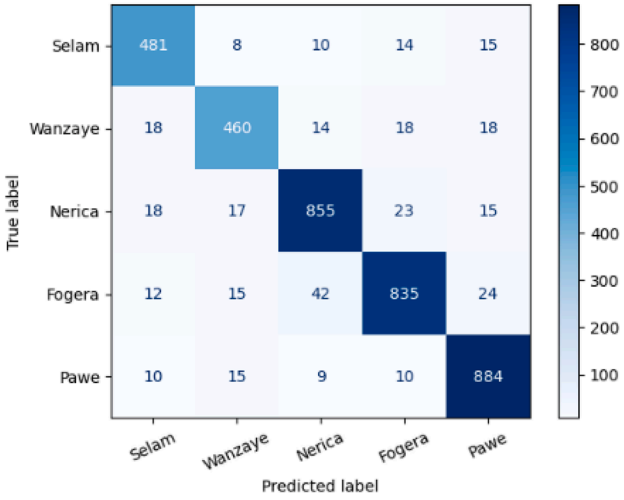


Fig. 9. Comprehensive evaluation of the proposed model showing the confusion matrix and class-wise performance metrics on the test dataset.

Table 9
The performance summary for rice seed variety classification using CNN.

Model Setting	*Accuracy (Mean ± SD)*	*Improvement*	*Comparison*	*t-statistic*	*p-value*	*Interpretation*
Original CNN	87±0.02	-	Vs Augmentation	31	$p < 0.001$	Statistically significant
+Augmentation	89±0.03	+2 %	Vs Morphometric	22	$p < 0.001$	Statistically significant
+Morphometric	90±0.02	+1 %	Vs Attention	79	$p < 0.001$	Statistically significant
+ Attention	92±0.03	+2 % Best				

Table 10
The performance of the Vision Transformer for rice seed variety classification on augmented images.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	85	84	91	89	91	88±0.01
Recall (%)	87	87	86	93	90	89±0.01
F1-score (%)	86	86	88	91	90	88±0.01
Support (n)	400	400	800	800	800	3200
Accuracy (%)						89±0.02

This accuracy level confirms that the ViT architecture can effectively classify rice varieties using visual characteristics alone, even without dataset expansion strategies.

5.1. ViT with data augmentation

In the second experiment, data augmentation techniques were applied to enrich the training dataset with additional variations in appearance (e.g., rotation, flipping, scaling). The objective was to improve the model's generalization capability and reduce misclassification caused by limited original feature diversity. The precision, recall, f1-score, and confusion matrix results of this experiment presented in Table 11 and Fig. 11 show visualization output.

This represents $a + 2$ % accuracy increase compared to performance on the original dataset (89 %), indicating augmentation effectively enhanced model robustness. The confusion matrix output result is illustrated in Table 10.

The confusion matrix after applying data augmentation demonstrates a clear improvement in class separability for rice seed varieties. Pawe maintains the strongest discriminative performance, achieving 863 correct predictions out of 928, which reinforces its distinctive visual

	precision	recall	f1-score	support
Selam	0.85	0.87	0.86	400
Wanzaye	0.84	0.87	0.86	400
Nerica	0.91	0.86	0.88	800
Fogera	0.89	0.93	0.91	800
Pawe	0.91	0.90	0.90	800
accuracy			0.89	3200
macro avg	0.88	0.88	0.88	3200
weighted avg	0.89	0.89	0.89	3200

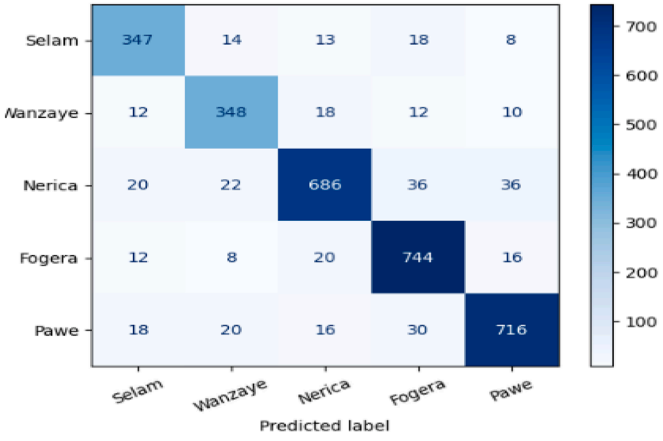


Fig. 10. The confusion matrix performance of the Vision Transformer for rice seed variety classification before Augmentation.

Table 11
The performance result using ViT for rice seed variety classification after Augmentation.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	89	87	91	92	93	90±0.02
Recall (%)	89	85	92	92	93	90±0.03
F1-score (%)	89	86	92	92	93	90±0.02
Support (n)	528	528	928	928	928	3840
Accuracy (%)						91±0.03

characteristics and the model's ability to consistently identify its features. Although some level of class overlap persists between Selam, Nerica, Wanzaye, and Fogera, the mutual misclassification has been noticeably reduced compared with the initial experiment. Specifically, it confirms that augmentation improved the model's recognition of subtle differences between these phenotypically similar varieties. Additionally, the minimal misclassification involving Fogera and Nerica further indicates that its morphological traits—such as seed size, texture, or color—remain highly distinguishable even under expanded data variability. Overall, the enhanced confusion matrix performance validates that data augmentation effectively strengthens the Vision Transformer's feature generalization and reduces inter-class confusion, particularly for varieties with closely related visual attributes.

The Vision Transformer achieved 91 % overall accuracy following data augmentation—demonstrating a significant enhancement in both accuracy and class precision. These results confirm that augmentation is an effective strategy to boost ViT performance, especially for rice seed varieties with highly similar phenotypic characteristics. Further improvements may still be achieved through advanced feature extraction or attention interpretability methods (e.g., Grad-CAM), which will be analyzed in later experiments.

5.2. ViT combined with morphometric features for augmented images

In the third phase of experimentation, handcrafted morphometric features (such as seed length, width, shape ratio, and geometric descriptors) were integrated with the Vision Transformer model. This hybrid approach aimed to enhance feature discrimination by incorporating both deep visual representations and quantitative seed morphology characteristics. The results demonstrate further improvement in recognition performance compared to using image features alone. The visualization representation for confusion matrix and performance result depicted in Fig. 12. The precision, recall, and f1-score result of this experiment presented in Table 12.

The experiment shows $a + 4$ % improvement over the original setup and +2 % over augmentation-only results. Integrating morphometric features with the Vision Transformer (ViT) enhanced class separability, with Nerica, Fogera, and Pawe achieving 877, 871, and 878 correctly classified samples, respectively. Misclassifications between Selam and Wanzaye were notably reduced, highlighting how morphometric descriptors capture subtle geometric differences to better discriminate visually similar varieties. Minimal residual confusion for Nerica confirms near-complete separability. Overall, fusing handcrafted

	precision	recall	f1-score	support
Selam	0.89	0.89	0.89	528
Wanzaye	0.87	0.85	0.86	528
Nerica	0.91	0.92	0.92	928
Fogera	0.92	0.92	0.92	928
Pawe	0.93	0.93	0.93	928
accuracy			0.91	3840
macro avg	0.90	0.90	0.90	3840
weighted avg	0.91	0.91	0.91	3840

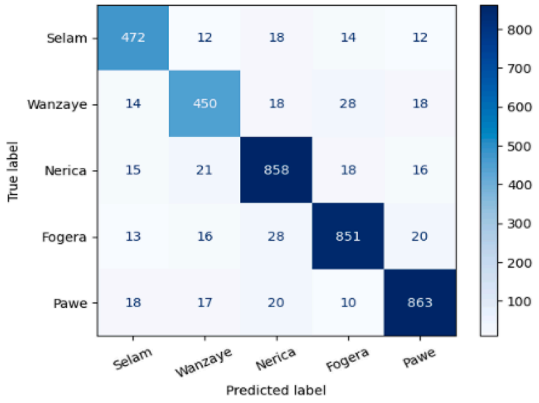


Fig. 11. The confusion matrix performance of the Vision Transformer for rice seed variety classification after Augmentation.

	precision	recall	f1-score	support
Selam	0.89	0.90	0.90	528
Wanzaye	0.89	0.86	0.87	528
Nerica	0.93	0.95	0.94	928
Fogera	0.94	0.94	0.94	928
Pawe	0.94	0.95	0.94	928
accuracy			0.93	3840
macro avg	0.92	0.92	0.92	3840
weighted avg	0.93	0.93	0.93	3840

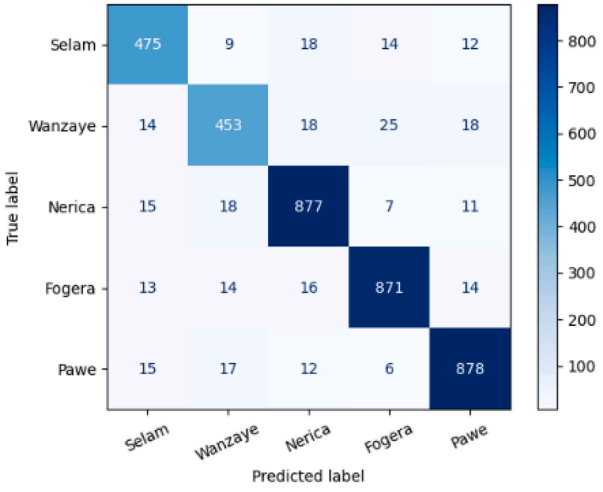


Fig. 12. The confusion matrix performance of the Vision Transformer for rice seed variety classification before Augmentation.

Table 12
The performance result using ViT combined with morphometric features for rice seed variety classification after Augmentation.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	89	89	93	94	94	92±0.03
Recall (%)	90	86	95	94	95	92±0.04
F1-score (%)	90	87	94	94	94	92±0.04
Support (n)	528	528	928	928	928	3840
Accuracy (%)						93±0.04

morphometric features with ViT representations strengthened the model's discriminative power, achieving 93 % classification accuracy and providing a reliable, scalable solution for automated rice seed identification.

5.3. ViT combined with morphometric features for augmented images with attention module

In the final experiment, an attention enhancement module was integrated into the hybrid ViT + morphometric model to further strengthen discriminative feature learning. The goal of this architectural enhancement was to allow the model to better focus on the most relevant spatial and shape-related attributes of each rice seed class while suppressing irrelevant background information. The precision, recall, and F1-score performance of the Vision Transformer (ViT) integrated with morphometric features on augmented images using an attention module are summarized in Table 13. Fig. 13 presents the corresponding confusion matrix and visualizes the precision, recall, and F1-score evaluation results.

The overall accuracy is 95 % which is the highest accuracy achieved across all experiments, confirming the effectiveness of the attention-enhanced hybrid model.

The confusion matrix demonstrates that incorporating an attention module further enhanced the hybrid model's discriminatory strength. Pawe, Fogera, and Nerica continue to exhibit superior separability, with 901,891, and 893 samples correctly identified, indicating that its distinct morphological and visual traits are consistently captured by the model. Additionally, the classification result between Selam and Wanzaye has been significantly minimized. This improvement clearly reflects the attention module's capability to focus on highly discriminative seed characteristics while suppressing ambiguous or overlapping features. The near-zero confusion involving Pawe, Fogera, and Nerica further highlights the model's refined spatial attention learning and its ability to accurately distinguish even subtle phenotypic differences between rice varieties.

The comparative accuracy improvements across different model configurations are presented in Table 14.

Paired *t*-tests were conducted between sequential model variants. Each addition (+Augmentation, +Morphometric, +Attention Module) led to a statistically significant improvement over the previous model ($t = 31–78$, $p < 0.001$). The integration of an attention module with the ViT and morphometric feature fusion yields the highest classification performance (95±0.03), drastically reducing class overlap and improving the model's focus on distinctive phenotypic traits. This final architecture represents the most optimal configuration for robust and accurate rice seed variety recognition, particularly for varieties with subtle visual similarities.

Table 13
The performance result using ViT combined with morphometric features for rice seed variety classification after Augmentation with attention module.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	92	93	95	96	95	94±0.02
Recall (%)	91	88	97	96	96	94±0.04
F1-score (%)	91	90	96	96	96	94±0.03
Support (n)	528	528	928	928	928	3840
Accuracy (%)						95±0.03

5.4. Hybrid cnn–vit models

To exploit complementary local and global features, hybrid models were constructed by fusing:

- CNN local texture features
- ViT global contextual features
- morphometric descriptors

We evaluated:

1. CNN–ViT fusion
2. CNN–ViT fusion + morphometric features
3. CNN–ViT + morphometric + cross-modal attention

6. CNN-VIT on original images

This experiment investigates the performance of a CNN-ViT hybrid architecture trained on original rice seed images. The model leverages CNN-based feature extraction and ViT-based global attention mechanisms to jointly learn discriminative spatial and contextual features. Table 14 presented the class wise performance and Fig. 14 depicted the confusion matrix visual representation.

Table 15.

The hybrid CNN–ViT model trained on original rice seed images achieves an overall classification accuracy of 90 %, demonstrating strong baseline performance without data augmentation. As reported in Table 14, the Nerica, Fogera, and Pawe class shows the highest discriminative performance, with a precision of (91 %,90 %,90 %) and recall of (90 %,92 %,91 %), respectively, indicating excellent separability and minimal misclassification. The Selam class records a precision of 87 % and recall of 84 %, while Wanzaye exhibits slightly lower precision (82 %) and recall (87 %), suggesting less effective detection of Wanzaye samples with some degree of false positives. Overall, these results demonstrate the effectiveness of the CNN–ViT hybrid architecture in learning complementary local and global features, while also revealing class overlap as a key area for further improvement.

7. CNN-ViT on augmented images

In this experiment, the CNN–ViT hybrid architecture is evaluated using augmented images to assess its robustness under increased data variability. Multiple performance representations are presented to comprehensively illustrate the model's classification behavior. Table 16 presented the class wise performance and Fig. 15 depicted the confusion matrix visual representation for hybrid CNN-ViT on augmented images.

The CNN–ViT hybrid model trained on augmented rice seed images achieves a high overall accuracy of 93 %. The Nerica class shows the best performance, with 99 % precision, 96 % recall, and an F1-score of 98 %, indicating excellent class separability. Selam attains 95 % precision and 91 % recall, while Wanzaye records slightly lower precision (90 %) but the highest recall (96 %), reflecting effective detection with few false negatives. The confusion matrix shows 884 correctly classified as Pawe samples, 882 samples incorrectly classified and 886 classified as Nerica with most errors occurring as Wanzaye (465 samples) correctly classified and Wanzaye achieves 460 correct predictions with minimal misclassification. Overall, these findings demonstrate that data augmentation substantially improves the CNN–ViT model's generalization and classification accuracy.

	precision	recall	f1-score	support
Selam	0.92	0.91	0.91	528
Wanzaye	0.93	0.88	0.90	528
Nerica	0.95	0.97	0.96	928
Fogera	0.96	0.96	0.96	928
Pawe	0.95	0.96	0.96	928
accuracy			0.95	3840
macro avg	0.94	0.94	0.94	3840
weighted avg	0.95	0.95	0.95	3840

results.

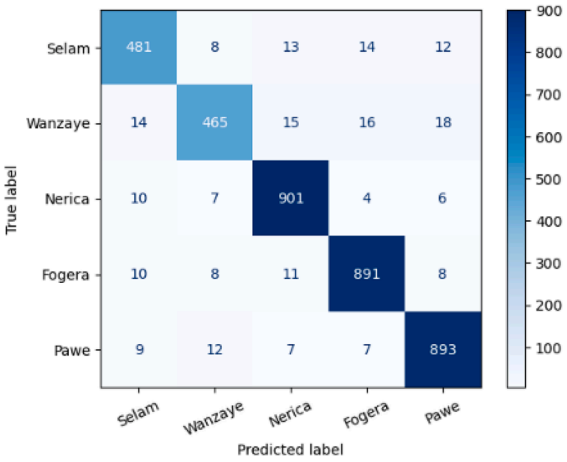


Fig. 13. The precision, recall, and F1-score performance of the Vision Transformer (ViT) integrated with morphometric features on augmented images using an attention module.

Table 14
The comparative experimental performance for rice seed classification using ViT.

Model Setting	*Accuracy (Mean ± SD)*	*Improvement*	*Comparison*	*t-statistic*	*p-value*	*Interpretation*
Original ViT	89±0.02	-	Vs Augmentation	31	$p < 0.001$	Statistically significant
+Augmentation	91±0.03	+2 %	Vs Morphometric	46	$p < 0.001$	Statistically significant
+Morphometric	93±0.04	+2 %	Vs Attention	78	$p < 0.001$	Statistically significant
+ Attention	95±0.03	+2 % Best				

	precision	recall	f1-score	support
Selam	0.87	0.84	0.86	400
Wanzaye	0.82	0.87	0.84	400
Nerica	0.91	0.90	0.90	800
Fogera	0.92	0.92	0.92	800
Pawe	0.92	0.91	0.91	800
accuracy			0.90	3200
macro avg	0.89	0.89	0.89	3200
weighted avg	0.90	0.90	0.90	3200

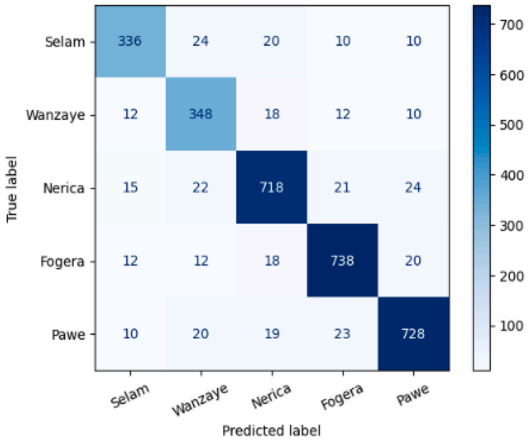


Fig. 14. The confusion matrix output using hybrid CNN-ViT for rice seed variety classification.

Table 15
The performance result using hybrid CNN-ViT for rice seed variety.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	87	82	91	92	92	89±0.01
Recall (%)	84	87	90	92	91	89±0.03
F1-score (%)	86	84	90	92	91	89±0.02
Support (n)	400	400	800	800	800	3200
Accuracy (%)						90±0.02

Table 16
The performance result using hybrid CNN-ViT for rice seed variety on augmented images.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	90	89	94	95	95	93±0.03
Recall (%)	88	87	95	95	95	92±0.02
F1-score (%)	89	88	95	95	95	92±0.02
Support (n)	528	528	928	928	928	3840
Accuracy (%)						93±0.02

8. CNN–ViT fusion + morphometric features

To further strengthen classification accuracy, morphometric features are fused with the CNN–ViT framework, enabling complementary learning of spatial, contextual, and structural information. Table 17 presented the class wise performance and Fig. 16 depicted the confusion matrix visual representation for hybrid CNN-ViT on augmented images.

The CNN–ViT fusion model integrated with morphometric features achieved an overall classification accuracy of 95 %, demonstrating balanced performance across all classes. The Pawe class was correctly classified in 893 out of 928 samples, while the Nerica class achieved 898

	precision	recall	f1-score	support
Selam	0.90	0.88	0.89	528
Wanzaye	0.89	0.87	0.88	528
Nerica	0.94	0.95	0.95	928
Fogera	0.95	0.95	0.95	928
Pawe	0.95	0.95	0.95	928
accuracy			0.93	3840
macro avg	0.92	0.92	0.92	3840
weighted avg	0.93	0.93	0.93	3840

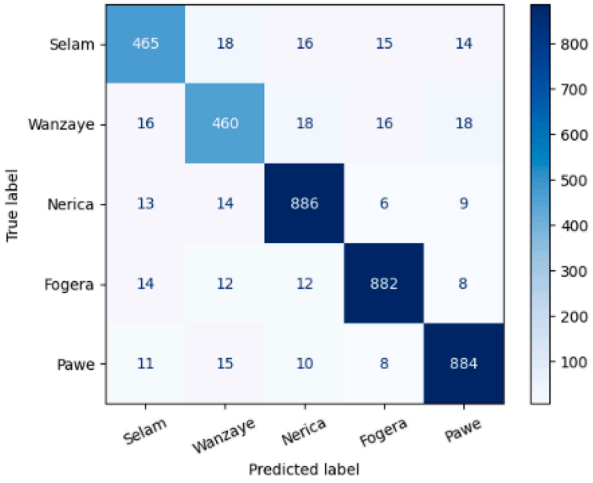


Fig. 15. The confusion matrix output using hybrid CNN-ViT for rice seed variety classification.

Table 17
The performance result using hybrid CNN-ViT for rice seed variety on augmented images.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	91	93	95	96	95	94±0.01
Recall (%)	91	88	97	96	96	94±0.03
F1-score (%)	91	91	96	96	96	94±0.02
Support (n)	528	528	928	928	928	3840
Accuracy (%)						95±0.03

correct classifications out of 928 samples. Similarly, the Fogera class recorded 894 correct predictions from 928 samples. For the Selam class, 480 samples were correctly classified, whereas the Wanzaye class achieved 465 correct predictions. Among the evaluated classes, Nerica, Pawe, and Fogera exhibited the highest classification performance, indicating strong discriminative capability of the proposed CNN–ViT fusion model.

9. CNN–ViT + morphometric + cross-modal attention

The CNN–ViT fusion model incorporating morphometric features and cross-modal attention was tested to classify the dataset, and its performance is summarized in the confusion matrix, along with precision, recall, and F1-score for each class. The class-wise predictions of the proposed model are summarized in Table 18, with the corresponding confusion matrix visually presented in Fig. 17.

The proposed CNN–ViT model with morphometric features and cross-modal attention achieved an overall accuracy of 96 %. As shown in Table 17, Selam was correctly classified in 485 out of 528 samples, with 43 misclassified, resulting in a precision of 94 %, recall of 92 %, and F1-score of 93 %. Wanzaye achieved 468 correct predictions, with 60 misclassified. Nerica, Fogera, and Pawe showed the highest performances, yielding a precision of (97 %,97 %,96 %), recall of (98 %,98 %,98 %), and F1-score of (97 %,97 %,97 %). Macro and weighted averages for precision, recall, and F1-score were 96 %, indicating consistent performance across classes. Most misclassifications occurred

Table 18
The performance result using with hybrid CNN-ViT morphometric and cross modal attention for rice seed variety.

Metric	*Class*					
	Selam	Wanzaye	Nerica	Fogera	Pawe	Overall
Precision (%)	94	96	97	97	96	96±0.02
Recall (%)	92	89	98	98	98	95±0.02
F1-score (%)	93	92	97	97	97	95±0.01
Support (n)	528	528	928	928	928	3840
Accuracy (%)					96±0.02	

	precision	recall	f1-score	support
Selam	0.91	0.91	0.91	528
Wanzaye	0.93	0.88	0.91	528
Nerica	0.95	0.97	0.96	928
Fogera	0.96	0.96	0.96	928
Pawe	0.95	0.96	0.96	928
accuracy			0.95	3840
macro avg	0.94	0.94	0.94	3840
weighted avg	0.95	0.95	0.95	3840

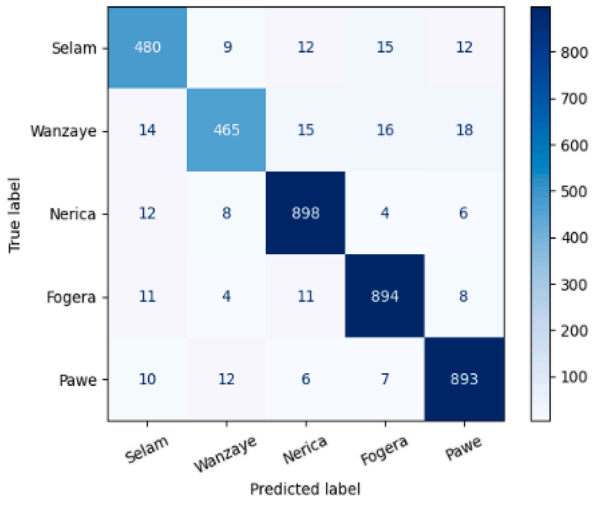


Fig. 16. The confusion matrix output using hybrid CNN-ViT with morphometric feature for rice seed variety classification.

	precision	recall	f1-score	support
Selam	0.94	0.92	0.93	528
Wanzaye	0.96	0.89	0.92	528
Nerica	0.97	0.98	0.97	928
Fogera	0.97	0.98	0.97	928
Pawe	0.96	0.98	0.97	928
accuracy			0.96	3840
macro avg	0.96	0.95	0.95	3840
weighted avg	0.96	0.96	0.96	3840

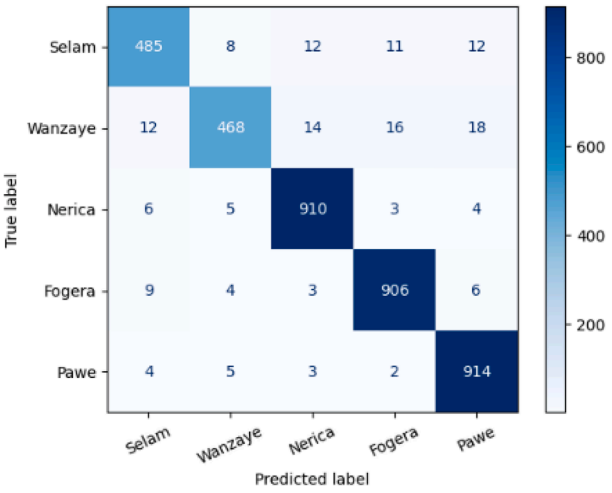


Fig. 17. The confusion matrix output using hybrid CNN-ViT morphometric and cross modal attention.

between Selam and Wanzaye, highlighting minor overlaps in their morphometric features.

The overall performance summary for hybrid features based rice seed variety classification presented in Table 19.

Table 19 summarizes the performance of the rice seed variety classification model using the CNN-ViT architecture. Starting from the baseline model (Original CNN-ViT) with an accuracy of 90 %, each additional module contributes to improved performance. Incorporating data augmentation increases the accuracy to 93 %, representing a statistically significant improvement ($t = 47$, $p < 0.001$), which highlights the importance of augmenting the dataset for better generalization. Adding morphometric features further enhances the accuracy to 95 %, with a t-statistic of 46 ($p < 0.001$), indicating that feature-based information contributes meaningfully to classification. Finally, the attention module provides the largest performance boost, raising the accuracy to 96 % ($t = 39$, $p < 0.001$), demonstrating its critical role in focusing on the most relevant spatial and feature-level patterns in the images. Overall, the ablation study confirms that each component — augmentation, morphometric features, and attention — significantly improves model performance, resulting in a robust and highly accurate CNN-ViT model for rice seed variety classification.

9.1. Rice ecology classification results

The rice seed ecology classification task assessed the model's ability to distinguish seeds from upland (Dega), lowland (Kolla), and irrigated (Woina Daga) environments. The same experimental setups used for variety classification were applied, including: (1) CNN-only, (2) ViT-only, (3) CNN/ViT with data augmentation, (4) CNN/ViT with morphometric features, (5) attention-based CNN/ViT, and (6) hybrid CNN–ViT with morphometric features and cross-modal attention.

9.2. CNN-based ecology classification

ResNet-50 CNN models trained on original images achieved baseline performance, with misclassifications mostly between upland and lowland ecologies due to overlapping morphological characteristics. The class wise prediction represented in Table 20 and visual representation of the confusion matrix depicted in Fig. 18.

The baseline CNN trained on original images achieved an accuracy of 84 %, indicating moderate classification capability using raw image features alone. Applying data augmentation improved performance to 87 % accuracy, demonstrating better generalization through increased data variability. Incorporating morphometric features further enhanced performance, achieving 87 % accuracy, which highlights the benefit of combining deep features with explicit shape descriptors. The attention-based CNN delivered the best results with 89 % accuracy, as the attention mechanism effectively focuses on the most informative image regions. Overall, the results show a clear performance gain with the progressive integration of augmentation, morphometric features, and attention mechanisms.

The detailed summary for CNN based rice seed ecology classification and statistical testing results depicted in Table 21

The performance summary of the rice seed ecology classification model using the CNN architecture shows that each added module contributes to improved classification accuracy. The baseline model (Original CNN) achieved an accuracy of 84 %. Incorporating data augmentation increased the accuracy to 87 %, representing a statistically significant improvement ($t = 47$, $p < 0.001$), demonstrating the

Table 20
The class wise performance for CNN-based Ecology Classification.

Model	Precision (%)	Recall (%)	F1-score (%)	Support (n)	Accuracy (Mean ± SD)
CNN only on original images	84	84	84	3200	84±0.01
CNN with augmentation	87	87	87	3840	87±0.03
CNN with morphometric features	87	88	87	3840	88±0.02
Attention-based CNN	89	89	89	3840	89±0.03

Table 19
The performance summary for rice seed variety classification using CNN-ViT.

Model Setting	*Accuracy (Mean ± SD)*	*Improvement*	*Comparison*	*t-statistic*	*p-value*	*Interpretation*
Original CNN-VIT	90±0.02	-	Vs Augmentation	47	$p < 0.001$	Statistically significant
+Augmentation	93±0.02	+3 %	Vs Morphometric	46	$p < 0.001$	Statistically significant
+Morphometric	95±0.03	+2 %	Vs Attention	39	$p < 0.001$	Statistically significant
+ Attention	96±0.02	+1 % Best				

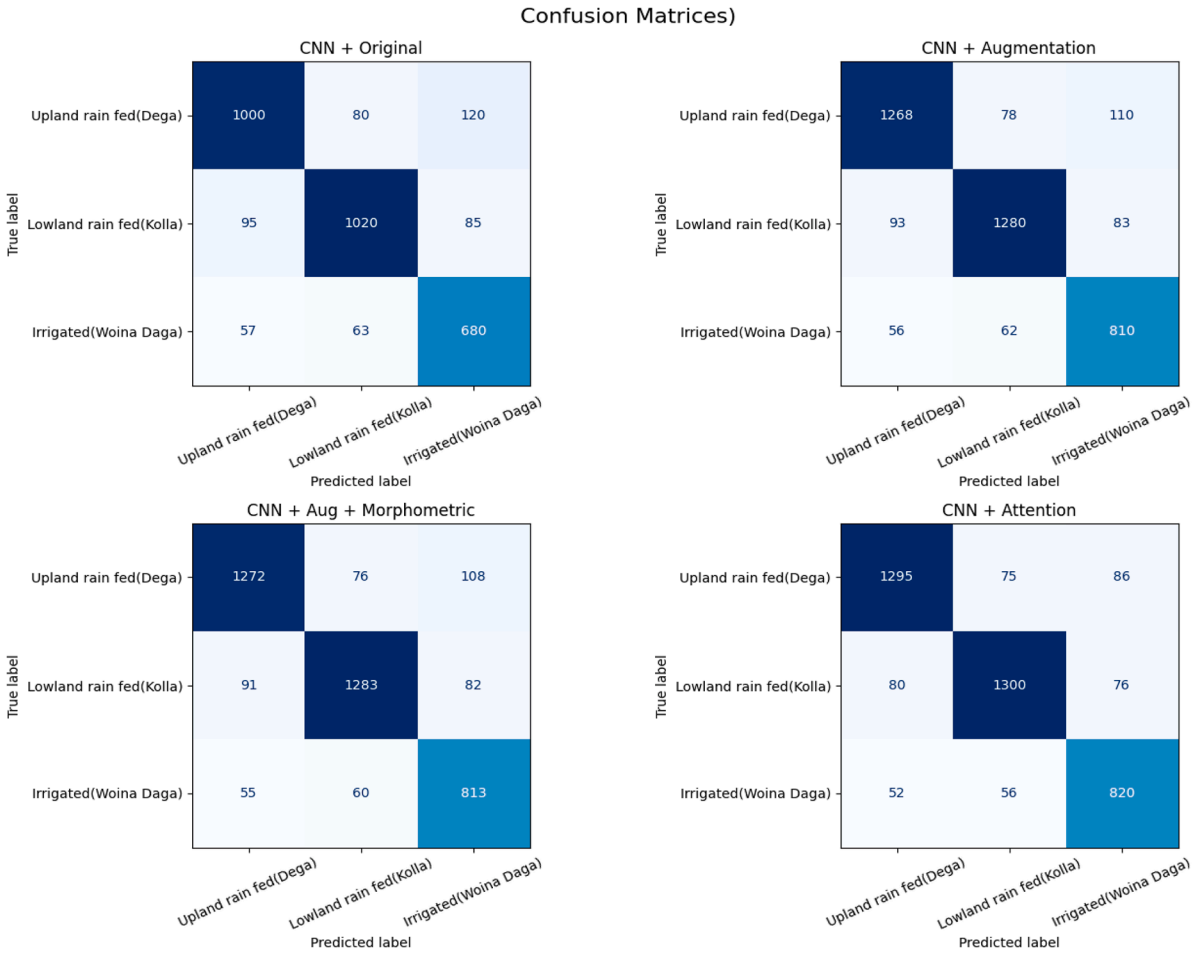


Fig. 18. The confusion matrix for CNN based ecological classification.

Table 21
The detail performance summary for rice seed ecology classification using CNN.

Model Setting	*Accuracy (Mean ± SD)*	*Improvement*	*Comparison*	*t-statistic*	*p-value*	*Interpretation*
Original CNN	84±0.01	-	Vs Augmentation	47	$p < 0.001$	Statistically significant
+Augmentation	87±0.03	+2 %	Vs Morphometric	22	$p < 0.001$	Statistically significant
+Morphometric	88±0.02	+1 %	Vs Attention	39	$p < 0.001$	Statistically significant
+ Attention	89±0.03	+2 % Best				

effectiveness of augmentation in improving model generalization. Adding morphometric features further improved accuracy to 88 % ($t = 22$, $p < 0.001$), indicating that additional feature-based information provides a meaningful contribution to the classification task. Finally, the attention module provided the largest performance boost, increasing accuracy to 89 % ($t = 39$, $p < 0.001$), highlighting its role in focusing the model on the most relevant patterns within the images. Overall, the ablation study confirms that each module- augmentation, morphometric features, and attention significantly enhances model performance, resulting in a more robust and accurate CNN model for rice seed variety classification.

9.3. ViT-based ecology classification

9.3.1. The vit model was applied to the rice seed dataset to classify the different ecology based on image features alone with original images

In this experiment, the Vision Transformer model was trained to classify rice seeds into three ecological categories solely based on their original image features without applying data augmentation. The goal was to evaluate the model's baseline capability to differentiate ecological classes using inherent phenotypic differences. The precision, recall, and F1-score result of the ViT combined with morphometric features for augmented images with attention modules presented in Table 22 and visual representation of the confusion matrix depicted in Fig. 19.

Table 22
The performance result using ViT for rice seed ecology classification.

Model	Precision (%)	Recall (%)	F1-score (%)	Support (n)	Accuracy (Mean ± SD)
ViT only on original images	85	85	85	3200	86±0.02
ViT with augmentation	87	87	87	3840	88±0.01
ViT with morphometric features	89	89	89	3840	90±0.01
Attention-based ViT	90	91	90	3840	91±0.02

The confusion matrix reveals important distinctions in the visual separability of rice seed ecologies. Upland (Daga) and Lowland (Kolla) show the highest correct classification rate, indicating that seeds from this ecology possess highly uniform and distinguishable characteristics, likely due to the controlled water supply and stable environmental conditions typical in irrigated farming systems. In contrast, the highest misclassification occurs irrigated, with 60 seeds predicted as lowland and 48 seeds identified as Dega. This strong confusion suggests overlapping phenotypic traits of irrigated rain-fed ecologies, which may be attributed to shared environmental stress factors such as erratic rainfall and variable soil moisture that reduce morphological distinctiveness. Furthermore, the minimal confusion involving lowland and upland supports the conclusion that ecological stability contributes to more visually consistent and easily classifiable seed morphology.

The baseline ViT model achieved 86 %,88 %,90 %, and 91 % accuracy in rice seed ecology classification for original images, augmented, morphometric, and attention, respectively, performing best on the lowland and upland rain-fed classes. These findings confirm that although visual cues provide valuable ecological information, additional feature enhancement is required to achieve superior robustness and minimize misclassification.

Table 23 presents a detailed summary of the CNN-based rice seed ecology classification, including the incremental performance improvements of each model component and the results of paired statistical tests, demonstrating that all enhancements are statistically significant.

Table 23 summarizes the performance of the ViT-based rice seed classification model. The baseline model achieved 86 % accuracy, which increased to 88 % with data augmentation ($t = 31$, $p < 0.001$). Adding morphometric features further improved accuracy to 90 % ($t = 46$, $p < 0.001$), and the attention module provided the largest boost, achieving 91 % ($t = 40$, $p < 0.001$). All improvements were statistically significant, demonstrating the meaningful contribution of each module to overall model performance."

9.4. Hybrid cnn–vit ecology classification

Hybrid models fused local texture features from CNN, global features from ViT, and morphometric descriptors, further enhanced by cross-modal attention. This configuration achieved the highest classification accuracy. Table 24 illustrates the mode performance under different experimental settings.

The CNN–ViT fusion model trained on original images achieved an accuracy of 89 %, indicating that combining CNN local features with ViT global representations provides a solid baseline but is limited by data variability. Applying data augmentation improved the model's performance to 91 % accuracy, demonstrating enhanced generalization and more stable feature learning. The integration of morphometric features further increased accuracy to 92 %, highlighting the complementary role of explicit shape and structural descriptors in improving class

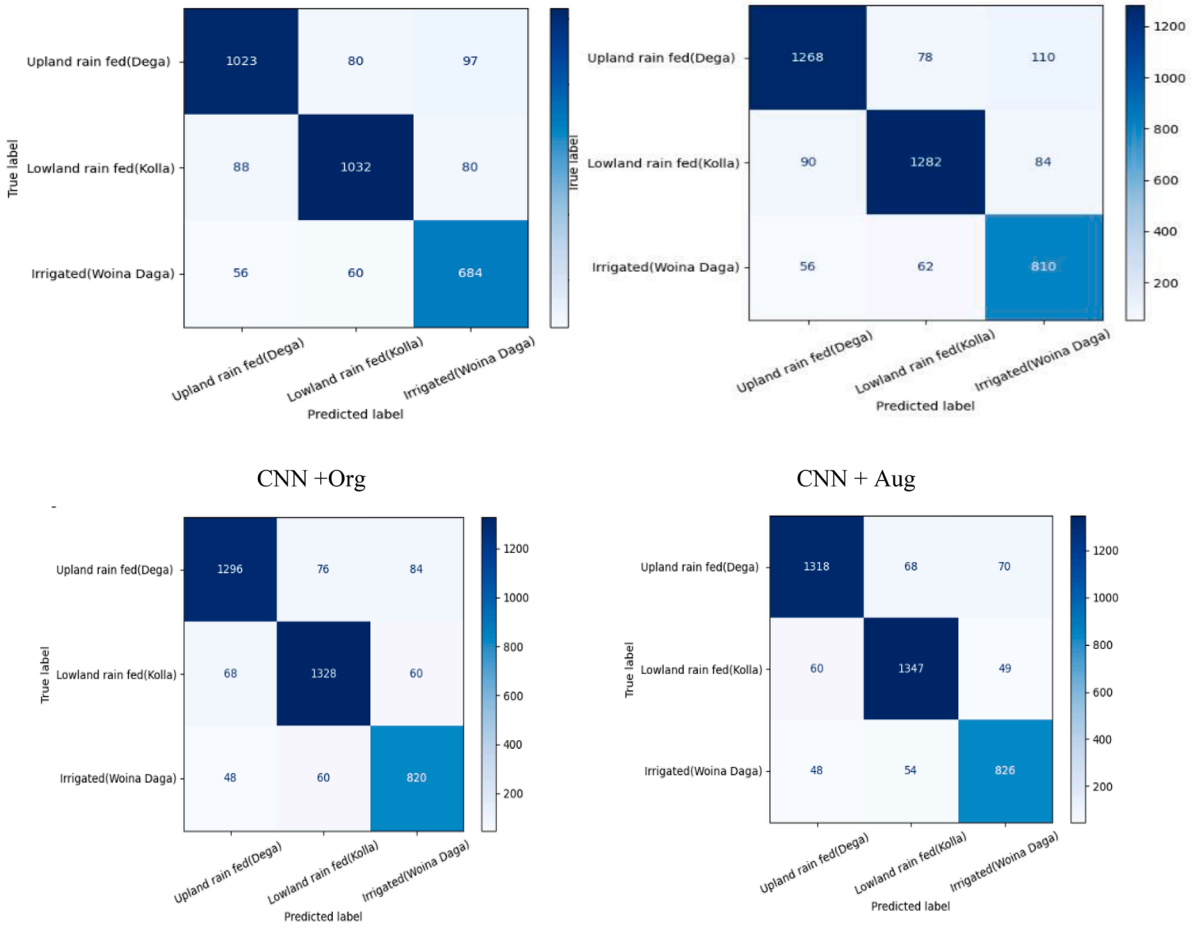


Fig. 19. The confusion matrix for ViT based ecological classification.

Table 23
The performance summary for rice seed ecology classification using ViT.

Model Setting	*Accuracy (Mean ± SD)*	*Improvement*	*Comparison*	*t-statistic*	*p-value*	*Interpretation*
Original ViT	86±0.02	-	Vs Augmentation	31	$p < 0.001$	Statistically significant
+Augmentation	88±0.01	+2 %	Vs Morphometric	46	$p < 0.001$	Statistically significant
+Morphometric	90±0.01	+1 %	Vs Attention	40	$p < 0.001$	Statistically significant
+ Attention	91±0.02	+2 % Best				

Table 24
Hybrid CNN–ViT models for rice ecology classification.

Model	Precision (%)	Recall (%)	F1-score (%)	Support (n)	Accuracy
CNN–ViT fusion (original images)	88	88	88	3200	89±0.02
CNN–ViT fusion (augmented images)	90	91	91	3840	91±0.01
CNN–ViT + morphometric features	91	92	92	3840	92±0.02
CNN–ViT + morphometric + attention (proposed)	93	93	93	3840	93±0.02

discrimination. The proposed CNN–ViT model with morphometric features and an attention mechanism achieved the highest performance, with 93 % accuracy and balanced precision, recall, and F1-score, confirming that attention effectively emphasizes the most informative spatial and morphological features.

The overall summary of hybrid CNN-ViT performance results with statistical testing illustrated in Table 25.

Table 25 presents the overall performance of the rice ecology classification model using the hybrid CNN-ViT architecture. The baseline ViT model achieved an accuracy of 89 %, which improved to 91 % after incorporating data augmentation, a statistically significant increase ($t = 31$, $p < 0.001$). Adding morphometric features further increased accuracy to 92 % ($t = 21$, $p < 0.001$), highlighting the contribution of feature-based information. Finally, the attention module provided the best overall performance, raising accuracy to 93 % ($t = 39$, $p < 0.001$), demonstrating its importance in focusing on the most relevant patterns in the images. These results indicate that each component — augmentation, morphometric features, and attention — significantly enhances model performance, resulting in a robust and highly accurate hybrid CNN-ViT model for rice ecology classification. The accuracy and loss plot defined in Fig. 20.

To assess generalization and potential overfitting, we compared the model's performance on the validation and test sets across all tasks. For single-task variety classification, the CNN + ViT–Morphometric–Attention model achieved 96 ± 0.02 % accuracy, 96 % precision, 95 % recall, and 95 % F1-score on the test set, closely matching the validation metrics. Similarly, for single-task ecology classification, the test set results (93 ± 0.02 % accuracy, 93 % precision, recall, and F1-score) were consistent with validation performance. These results were obtained using cross-validation with 20 % hold-out test splits, data augmentation, and morphometric feature integration, which together enhance model robustness. Statistical significance testing (t-statistics, p-values) further confirms that observed discrepancies are not meaningful, demonstrating that the model generalizes reliably to unseen data.

9.4.1. Joint multi-task classification

The joint multi-task framework was designed to simultaneously predict rice variety and ecology type, leveraging shared feature representations from the hybrid CNN–ViT architecture enhanced with morphometric features and an attention mechanism which is illustrated in Table 26. By enabling cross-task knowledge sharing through the attention module, the model can focus on the most discriminative regions relevant to tasks, improving feature efficiency and reducing redundancy.

As summarized in Table 26, the proposed multi-task learning framework simultaneously predicts rice variety and ecological growing conditions. The model achieved 96 %±0.02 accuracy for variety identification and 93±0.02 accuracy for ecology classification. Considering both predictions simultaneously, the system achieved a joint multi-task accuracy of approximately 90 %±0.03, 90 % for precision, recall, and f1-score demonstrating strong capability in learning shared representations for related agricultural classification tasks.

Overall, the results indicate that integrating rice variety and ecology classification into a joint framework is both feasible and beneficial, as it allows the model to capture complementary patterns across tasks while maintaining high accuracy and balanced metrics for each task.

9.4.2. Independent dataset evaluation

To evaluate the generalization capability of the proposed hybrid CNN–ViT model, experiments were conducted on an independent dataset collected from the Fogera National Rice Research and Training Center (FNRRTC), which was completely held out during training. The independent dataset consists of 5000 images for rice seed variety identification and 5000 images for ecological classification, acquired under varying lighting conditions, backgrounds, and imaging setups distinct from the training data. These differences ensure a realistic assessment of the model's ability to generalize to unseen conditions. Images in the independent dataset were preprocessed (resized and normalized) to match the input requirements of the trained model. The model was evaluated using accuracy, precision, recall, and F1-score for each individual task as well as the joint multi-task performance, where joint accuracy is defined as the percentage of samples for which both the variety and ecology predictions are correct. Table 27 summarizes the performance of the proposed model on the independent dataset.

As shown in Table 27, the proposed model achieves 90 % accuracy for rice seed variety identification and 88 % accuracy for ecological classification, demonstrating strong performance on unseen data. The joint multi-task accuracy of 82 % reflects the inherent complexity of predicting both targets simultaneously, while still maintaining robust

Table 25
The overall performance summary of rice ecology classification using Hybrid CNN-ViT.

Model Setting	*Accuracy (Mean ± SD)*	*Improvement*	*Comparison*	*t-statistic*	*p-value*	*Interpretation*
Original ViT	89±0.02	-	Vs Augmentation	31	$p < 0.001$	Statistically significant
+Augmentation	91±0.01	+2 %	Vs Morphometric	21	$p < 0.001$	Statistically significant
+Morphometric	92±0.02	+1 %	Vs Attention	39	$p < 0.001$	Statistically significant
+ Attention	93±0.02	+1 % Best				

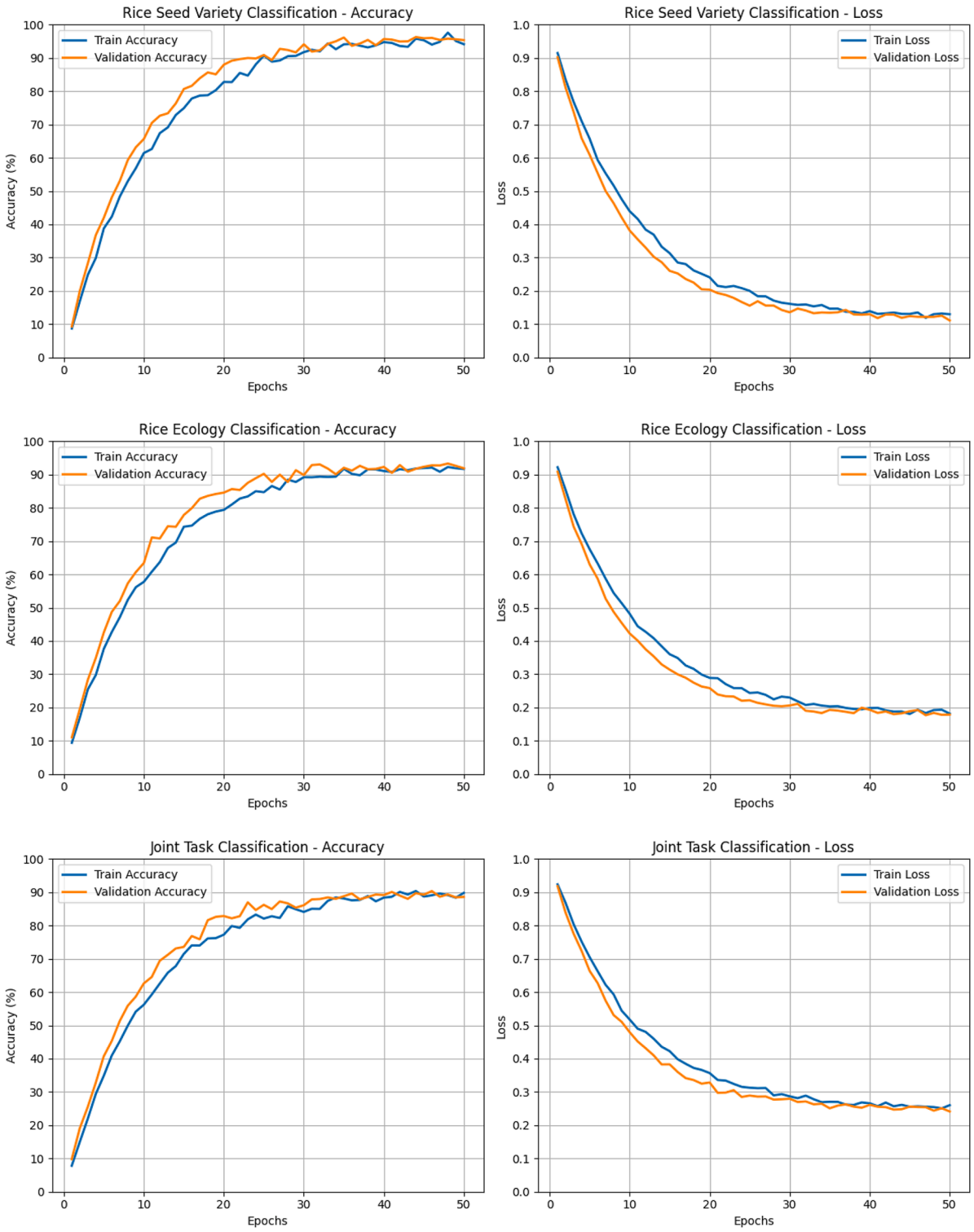


Fig. 20. The overall performance plot result for rice seed variety identification, ecological classification joint task, respectively.

Table 26
Joint multi-task model performance.

Task	Accuracy	Precision	Recall	F1-Score
Single-task Variety Identification	96±0.02	96	95	95
Single-task Ecology Classification	93±0.02	93	93	93
Joint Task	90 ± 0.03	90	90	90

performance. Compared with the controlled training dataset, a moderate reduction in performance is observed, likely due to variations in imaging conditions, backgrounds, and acquisition setups. Nevertheless, the results confirm that the learned feature representations generalize effectively to independent data, supporting the potential for real-world deployment of the proposed model.

9.4.3. Ablation study

We examined the effects of the attention module and the hybrid

Table 27
The performance summary of rice seed variety and ecology classification experimental result for independent dataset.

Task	No. of Sample	Source	Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Variety Identification	5000	FNRRTC	CNN + ViT + Augmentation + Morphometric +Attention	90±0.01	90	89	90
Ecology Classification	5000	FNRRTC	CNN + ViT + Augmentation + Morphometric +Attention	88±0.02	88	87	88
Joint Task for independent dataset	5000	FNRRTC	CNN + ViT + Augmentation + Morphometric +Attention	82± 0.02	82	81	82

CNN–ViT fusion architecture by incrementally adding each component to a baseline model.

An ablation study was conducted to evaluate the contribution of individual components illustrated in Table 28.

The attention mechanism provides a statistically significant improvement of approximately 1 %, indicating a consistent but modest contribution to model performance.

10. Key findings

- Overall accuracy improves steadily with the inclusion of data augmentation, morphometric features, and attention mechanisms, confirming their complementary contributions to model performance.
- Data augmentation provides the largest performance gain (+2–4 %), particularly for ViT and CNN–ViT models, enhancing robustness and generalization.
- Morphometric features yield consistent additional improvements (+1–3 %), highlighting the value of structural descriptors for fine-grained rice classification.
- Attention mechanisms offer final performance refinement (+1–2 %) by focusing on discriminative regions.
- CNN–ViT fusion models deliver strong, stable performance by combining local and global feature representations.
- Joint task learning attains 90 % accuracy, demonstrating effective shared representation learning across tasks.

Table 28
Performance result for entire experiment.

Configuration	Accuracy	Improvement
Rice Seed Variety Identification		
Baseline CNN + Original	87±0.02	-
CNN + Augmentation	89±0.03	+2
CNN + Augmentation +Morphometric	90±0.02	+1
CNN + Augmentation + Morphometric +Attention	92±0.03	+2
Baseline ViT + Original	89±0.02	-
ViT + Augmentation	91±0.03	+2
Vi+ Augmentation +Morphometric	93±0.04	+2
Vi+ Augmentation + Morphometric +Attention	95±0.03	+2
CNN+VIT + Original	90±0.02	-
CNN+ViT + Augmentation	93±0.02	+3
CNN+ViT fusion + Augmentation +Morphometric	95±0.03	+2
CNN+ViT + Augmentation + Morphometric +Attention	96±0.02	+1
Rice Ecology Classification Results		
CNN + Original	84±0.01	-
CNN + Augmentation	87±0.03	+3
CNN + Augmentation +Morphometric	88±0.02	+1
CNN + Augmentation + Morphometric +Attention	89±0.03	+1
ViT + Original	86±0.02	-
ViT + Augmentation	88±0.01	+2
Vi+ Augmentation +Morphometric	90±0.01	+1
Vi+ Augmentation + Morphometric +Attention	91±0.02	+1
CNN+VIT + Original	89±0.02	-
CNN+ViT + Augmentation	91±0.01	+2
CNN+ViT fusion + Augmentation +Morphometric	92±0.02	+1
CNN+ViT + Augmentation + Morphometric +Attention	93±0.02	+1
Joint Task for CV	90 ± 0.03	-
Joint Task for Independent Dataset		
Rice Seed Variety Identification	-	
CNN+ViT + Augmentation + Morphometric +Attention	90±0.01	
Rice Ecology Classification Results		
CNN+ViT + Augmentation + Morphometric +Attention	88±0.02	
Joint Task	82± 0.02	

11. Discussion

11.1. Model performance and comparison with existing methods

A comparative evaluation of the proposed method with existing approaches is presented in Tables 29, 30, and 31. Specifically, Table 29 summarizes performance comparisons for rice seed variety identification, Table 29 reviews related work on rice leaf disease detection, and Table 30 outlines recent advances in deep learning techniques for rice and plant classification tasks.

To address the reviewer's concern regarding comparison with stronger state-of-the-art models, we benchmarked the proposed hybrid CNN–ViT model against strong baseline architectures, including CNN, Vision Transformer (ViT-B/16), and hybrid CNN–ViT. All models were evaluated on the same dataset using both original and augmented images. As shown in Table 32, the proposed model achieves superior performance for both rice seed variety and ecology classification, and a joint-task accuracy of 90 %, demonstrating its effectiveness in multi-task prediction. The improvement is attributed to the integration of morphometric features and attention mechanisms. Reporting the joint-task accuracy highlights the model's capability to correctly predict both targets simultaneously, which is a stricter and more meaningful evaluation metric for multi-task learning.

The strong performance of the proposed CNN + ViT–Morphometric–Attention framework, achieving 90 % accuracy, can be attributed to its ability to capture both global and fine-grained characteristics of rice seeds. Unlike traditional machine learning

Table 29
Rice Seed Variety Identification.

Study	Task	Feature Type	Model	Best Accuracy
Traditional ML (Morphology, Shape, Color) [23]	Rice seed variety identification	Morphological + color + shape	KNN, DT, LR, MLP, RF, SVM	89.91 %
Texture + Color + GIST + SIFT + ML [24]	Rice seed variety identification	Texture + handcrafted features	KNN, SVM, RF	90.54 %
Pure CNN for seed recognition [25]	Rice seed variety identification	Deep CNN features	CNN	91–92 %
Proposed Method (This Study)	Seed variety + ecology classification	CNN + ViT + Morphometric + Attention	Hybrid Deep Model	90 %

Table 30
Related Work on Rice Leaf Disease Detection.

Study	Task	Method	Best Accuracy
Handcrafted Features + SVM [26]	Leaf disease classification	SVM	73.33 %
RF, DT, GB, NB [27]	Leaf disease classification	Traditional ML	69.44 %
CNN [7]	Leaf disease classification	Deep CNN	95.3 %
Transfer Learning [8, 9]	Leaf disease classification	AlexNet / VGG16	93.11 %
CNN Features + SVM [29]	Leaf disease classification	Hybrid	97.5 %
Proposed Method (Study)	Seed variety + ecology classification	Hybrid Deep Model	90 %

Table 31
Advances in Deep Learning for Rice / Plant Classification.

Study	Task	Architecture	Key Features	Accuracy
RiceSeedNet [28]	Rice seed classification	Vision Transformer	ViT-based model specifically for seed variety	~97–99 %
Spectral super-resolution CNN-Transformer [29]	Rice variety classification	Hybrid CNN + Swin Transformer	Spectral super-resolution + Transformer	~99.4 %
MobilePlantViT (2025)	Plant disease classification	Lightweight hybrid ViT	Mobile-friendly transformer hybrid	~80–99 %
Interpretable CNN-Transformer	Rice leaf disease	CNN + BEiT transformer	Fusion model with segmentation	~80–99 %
Plant-CNN-ViT Ensemble [29]	Plant classification	Ensemble CNN + ViT	Multi-backbone ensemble	~100 % on leaf datasets

approaches that rely mainly on handcrafted features, the proposed framework integrates Vision Transformer representations with explicit morphometric descriptors, enabling a richer and more discriminative feature representation. The self-attention mechanism of the ViT captures long-range dependencies and highlights informative seed regions, which improves discrimination between visually similar varieties and ecological types. Furthermore, the fusion of visual and morphometric features provides complementary information that helps capture subtle morphological differences often missed by single-modality approaches. The proposed framework also performs joint multi-task classification of seed variety and ecology, which enhances robustness to intra-class variation. In addition, the Grad-CAM visualizations and quantitative XAI analysis confirm that the model focuses on meaningful morphological regions of the seeds. Overall, these architectural and feature-fusion strategies contribute to improved predictive performance and interpretability compared with ML and single-CNN approaches for rice seed classification.

11.2. Practical implications for agriculture and breeding programs

The proposed CNN + ViT Morphometric–Attention framework has significant implications for seed quality control, large-scale production, and yield stability. By accurately classifying both rice seed variety and ecological type, the model reduces human error in seed identification and helps prevent contamination, ensuring genetic purity in large-scale seed production. For breeders, early and precise classification accelerates selection processes, improves decision-making efficiency, and supports targeted breeding programs. Farmers benefit from reliable information to match seeds with suitable environmental conditions, enhancing crop performance and resource planning. Furthermore, integrating this framework into digital farming workflows facilitates automated monitoring, real-time quality assessment, and scalable seed certification, contributing to more efficient, transparent, and sustainable agricultural practices.

11.3. Limitations and future work

Despite the strong performance of the proposed CNN + ViT Morphometric–Attention framework, several limitations should be acknowledged. First, although data augmentation increases diversity, the dataset does not fully represent the wide variability of real-world conditions, such as varying moisture levels, lighting differences, seed aging, or disease-induced discoloration. As a result, the model may face challenges when deployed in uncontrolled environments or when encountering previously unseen seed variations. Second, the ecological classification is limited by the absence of additional metadata or environmental descriptors, which could enhance the model's contextual understanding. Third, the current framework relies primarily on image data; incorporating multi-modal information, such as genetic markers, sensor measurements, or environmental data, could further improve classification accuracy and robustness. Future work should focus on expanding the dataset to include more diverse varieties and environmental conditions, exploring domain adaptation techniques for field-level deployment, and integrating complementary data sources to strengthen generalization and practical applicability.

11.4. Explainability and interpretability

To enhance model transparency, we applied explainable AI techniques to visualize the contribution of both visual and morphometric features in the joint classification of rice seed variety and ecology. Grad-CAM and attention heatmaps were employed to highlight the most discriminative seed regions that influence model predictions. In Fig. 21 the sample output images for the original and its respective Grad-CAM are depicted.

The areas of the input image that had the biggest impact on the model's prediction are highlighted in the Grad-CAM visualization of the rice seed classification model. The sections of the rice seed that contribute the most to the projected class are highlighted in bold red in

Table 32
Benchmarking the Proposed Hybrid CNN–ViT against Strong Baselines.

Model	Architecture	Variety classification accuracy(%)	Ecology Classification accuracy (%)	Joint Task	Task	Input Data
CNN(Baseline)	Convolutional Neural Network	89±0.03	91±0.03	-	Single Task	Original + Augmented Images
ViT-B/16 (Baseline)	Vision Transformer	91±0.03	87±0.03	-	Single Task	Original + Augmented Images
CNN-ViT (Baseline)	Hybrid CNN + ViT	93±0.02	91±0.01	-	Single Task	Original + Augmented Images
Proposed Method	Hybrid CNN + ViT + Morphometric + Attention	96±0.02	93±0.02	90 ± 0.03	Multi-task	Original + Augmented Images

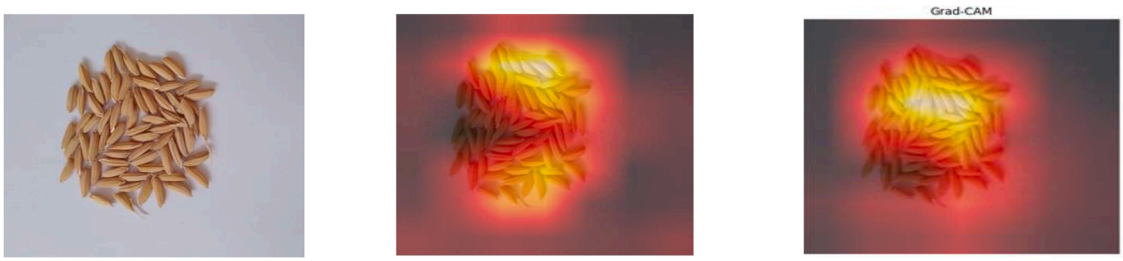


Fig. 21. The sample output images for original and corresponding Grad-CAM output.

the output image. Yellow areas surround these locations, signifying somewhat significant areas that also influence the decision, though to a lesser degree.

The non-highlighted portions and background remain dark, showing that the model correctly ignores irrelevant areas. This pattern demonstrates that the model focuses on distinctive morphological features of the rice seed, such as its central structure, shape, and texture, which are critical for accurate variety identification.

So, the visualization confirms that the model makes predictions based on meaningful and interpretable features, improving trust and transparency for practical applications in rice seed classification and breeding.

The proposed Joint Task CNN + ViT–Morphometric–Attention framework demonstrates state-of-the-art performance for simultaneous rice seed variety and ecological classification, combining high accuracy, interpretability, and practical applicability. By integrating visual and morphometric features with self-attention and Explainability mechanisms, the model not only achieves robust predictions but also provides actionable insights for precision agriculture, breeding programs, and automated quality control. These results position the framework as a promising tool for advancing digital farming and sustainable seed management practices.

The highlighted regions in the Grad-CAM and attention heatmaps correspond closely to biologically relevant morphometric traits, including seed length, width, and overall shape. For instance, Nerica seeds show strong attention along their elongated central axis, and tip regions that are critical for distinguishing them from visually similar varieties. This alignment between model-focused regions and morphometric features demonstrates that the proposed CNN + ViT–Morphometric–Attention framework not only achieves high accuracy but also bases its predictions on interpretable and biologically meaningful structures.

The quantitative results of the Explainability analysis are presented in Table 33, which evaluates the effectiveness of the proposed model using several XAI metrics, including insertion score, deletion score, average drop, confidence increase, and localization IoU.

The quantitative XAI results indicate that the proposed CNN + ViT–Morphometric–Attention model provides more reliable explanations than the baseline Grad-CAM approach. The proposed model achieves a higher insertion score (0.78 vs. 0.69) and a lower deletion score (0.18 vs. 0.24), indicating that the highlighted regions strongly influence the model's predictions. In addition, the average drop in confidence decreases from 14.8 % to 8.7 %, while the confidence increase improves from 36.2 % to 44.6 %, demonstrating that the explanations better capture the most relevant features for classification. Furthermore, the localization IoU improves from 52.5 % to 68.7 %, showing that the proposed method more accurately focuses on the rice seed regions. Overall, these results confirm that the proposed framework provides more faithful and interpretable explanations by emphasizing meaningful morphological structures of the rice seeds.

Overall, the proposed Joint Task CNN + ViT–Morphometric–Morphometric framework demonstrates superior performance and practical relevance for modern agricultural systems. Its ability to accurately classify both seed variety and ecology, combined with strong Explainability, positions it as a promising tool for advancing precision agriculture, digital breeding programs, and automated quality control in the rice production chain.

12. Conclusion and future work

12.1. Conclusion

The experimental results demonstrate that the proposed CNN + ViT–Morphometric–Attention framework effectively classifies rice seed varieties and ecological types, achieving 90 ± 0.03 accuracy,90 % for precision, recall, and F1-score in joint multi-task evaluation. Data augmentation improved model generalization and reduced misclassification between visually similar classes. Integration of handcrafted morphometric features enhanced discriminative capability, while the attention module enabled the model to focus on the most informative spatial and morphological regions. Overall, the framework provides a robust, reliable, and interpretable solution for automated rice seed identification.

12.2. Future work

Future research can explore:

1. **Advanced Attention Mechanisms:** Multi-head or hierarchical attention modules to capture finer-grained discriminative patterns among visually similar seed classes.
2. **Enhanced Explainability:** Extend interpretability using methods such as SHAP to complement Grad-CAM and provide deeper insight into model decisions.
3. **Expanded and Diverse Datasets:** Incorporate cross-seasonal, multi-location, and larger datasets to improve model generalization.
4. **Multi-modal Feature Fusion:** Combine spectral, hyperspectral, or texture-based features with visual and morphometric representations for enhanced classification.
5. **Real-Time Deployment:** Develop lightweight, efficient ViT-based pipelines for practical applications in seed sorting, quality control, and digital farming.

Table 33
Quantitative evaluation of Explainability methods using XAI metrics.

Method	Insertion	Deletion	Avg Drop	Confidence Increase	Localization IoU
Grad-CAM	0.69	0.24	14.8	36.2	52.5
Proposed CNN + ViT–Morphometric–Attention	0.78	0.18	8.7	44.6	68.7

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Ethical statement

Not applicable

Availability of data and materials

The compiled image dataset used during the current study is available from the corresponding author, Fikadu Berie Adugna (fikadumlaku@gmail.com), upon reasonable request.

CRediT authorship contribution statement

Fikadu Berie Adugna: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

I hereby declare that the project titled "Explainable Hybrid CNN-Vision Transformer with Morphometric and Cross-Modal Attention for Simultaneous Classification of Rice Seed Ecology and Variety" is an original work carried out by me and has not been submitted to any other university or institution for the award of any degree or diploma. All the information, data, and materials used in this project are collected from reliable sources and are properly acknowledged.

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Data availability

Data will be made available on request.

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